

Extended Area Blackbody System SR-800N

Installation and Operation Manual

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Preface

Scope of This Manual

This manual provides all the required information for installing, operating, and maintaining the SR-800N Extended Area Blackbody System. The manual is intended for use by operator-level personnel.

Contents of This Manual

The manual includes the following chapters:

Chapter 1: [Introduction & General information](#)

Introduction to the SR-800N Extended Area Blackbody, providing the SR-800N main features, general technical data, and specifications.

Chapter 2: [Description](#)

Description of the structure and assemblies of the SR-800N.

Chapter 3: [Handling & Installation](#)

Instructions for transporting, storing, unpacking, and installing the SR-800N.

Chapter 4: [Operation](#)

Detailed operation instructions for the SR-800N.

Chapter 5: [Maintenance](#)

Maintenance and troubleshooting information for the SR-800N.

Chapter 6: [Communication](#)

Information regarding remote access to the SR-800N.

Chapter 7: [Appendix A – Targets Wheel](#)

Description of the SR-800N Targets wheel construction and calibration procedure.

Chapter 8: [Index](#)

A list of keywords to be found throughout this manual Provides, in alphabetical order.

List of Updates (Effective Pages)

No.	Date	Description	Page Number
A	April 2016	New Release	All

Table of Contents

1.	INTRODUCTION & GENERAL INFORMATION	1-1
1.1	SR-800N Extended Area Blackbody System Overview	1-1
1.2	SR-800N Main Features	1-2
1.3	General Specifications.....	1-3
1.3.1	ControlMaster Mechanical specification.....	1-3
1.3.2	ControlMaster Electrical specification	1-3
1.3.3	SR-800N performance (common for all types of Blackbody heads).	1-3
2.	DESCRIPTION	2-1
2.1	SR-800N Block Diagram	2-1
2.2	SR-800N Main Components	2-2
2.3	Blackbody Head Overview	2-4
2.3.1	Blackbody Head Structure.....	2-4
2.3.2	Absolute Mode Blackbody Head	2-5
2.3.3	Differential Mode Blackbody Head	2-5
2.3.4	LT Blackbody Head Structure.....	2-6
2.4	ControlMaster Overview	2-8
2.4.1	ControlMaster Components	2-8
2.4.2	Controls & Indicators	2-8
2.4.3	Connectors.....	2-9
2.4.4	RS232 Connector Pinout	2-10
2.4.5	USB Jack Pinout.....	2-10
2.4.6	LAN Jack Pinout	2-11
2.4.7	I/O Connector Pinout (TTL Option)	2-11
2.4.8	I/O Connector Pinout (Galvanic Isolation Option)	2-12
2.4.9	I/O Connector Pinout (Encoder Feedback Option)	2-12
2.5	SR-800N Cables set.....	2-14
2.6	SR-800N Main Options.....	2-15
2.6.1	Motorized Axes	2-15
2.6.2	Encoder feedback	2-15
2.6.3	Special I/Os	2-15
2.6.4	Extended Temperature Range	2-15
2.6.5	Low Temperature Blackbody Heads.....	2-15
2.6.6	Wide Temperature Range.....	2-15
2.6.7	High Emissivity Emitter	2-15
2.6.8	Emissivity Compensation.....	2-15
2.6.9	High Uniformity	2-15
3.	HANDLING & INSTALLATION	3-16

3.1	Transportation	3-16
3.2	Storage.....	3-16
3.3	Installation	3-16
3.3.1	Unpacking	3-16
3.3.2	Installation Configurations	3-17
3.3.3	Low temperature (LT) Blackbody Head Installation	3-19
3.4	Removal.....	3-24
3.4.1	Desktop (stand alone) Configuration	3-24
3.4.2	19" rack Configuration	3-24
3.4.3	LT Blackbody Head	3-24
4.	OPERATION.....	4-26
4.1	Powering-on the SR-800N	4-26
4.2	Turning-Off the SR-800N	4-26
4.3	Operation Procedures	4-27
4.3.1	General navigation buttons.....	4-27
4.3.2	General Controls.....	4-28
4.3.3	General Data Entry Screens	4-29
4.3.4	Absolute Mode Screen	4-31
4.3.5	Differential Mode Screen	4-32
4.3.6	Setting the Temperature	4-33
4.3.7	Preset functions	4-34
4.3.8	Selecting an Axis item (Target, Filter or Source)	4-35
4.3.9	I/O Control Screen.....	4-37
4.3.10	'Info' screen.....	4-38
4.3.11	LT Blackbody Head Operation	4-40
4.4	Configuration Procedures.....	4-41
4.4.1	Main Menu screen	4-41
4.4.2	Preset Configuration screen.....	4-50
4.4.3	Axis Setting screen.....	4-52
4.4.4	Axis item Configuration screen.....	4-58
4.4.5	Direct Operation screen.....	4-63
5.	MAINTENANCE	5-64
5.1	Preventive/Periodic Maintenance Tasks	5-64
5.1.1	Visual Inspection.....	5-64
5.1.2	Cleaning.....	5-64
5.1.3	Calibration	5-64
5.2	Troubleshooting.....	5-65
5.2.1	General Failures	5-65

5.2.2	Failures screen	5-66
5.3	Fuse Replacement.....	5-69
5.3.1	Procedure.....	5-69
5.4	Temperature sensors Calibration methods	5-70
5.4.1	Using a standalone calibration instrument	5-70
5.4.2	Using a removable temperature sensor	5-70
5.5	Temperature Sensors Maintenance	5-72
5.5.1	4 ÷ 20" Blackbody Heads Emitter Sensor Replacement	5-72
5.5.2	4 ÷ 20" Blackbody Heads Target Sensor Replacement.....	5-76
5.5.3	2" Blackbody Head Emitter Sensor Replacement.....	5-77
5.5.4	2" Blackbody Head Target Sensor Replacement	5-79
5.5.5	Updating the Calibration Data.....	5-80
5.6	Updating the Software Version	5-83
5.6.1	Upgrade screen	5-83
6.	COMMUNICATION CHANNELS.....	6-1
6.1	Communication Ports description	6-1
6.1.1	RS232 serial port	6-1
6.1.2	Ethernet	6-1
6.2	Remote Operation.....	6-3
6.2.1	Overview	6-3
6.2.2	Communication Protocol.....	6-3
6.2.3	Remote Operation Application.....	6-3
7.	APPENDIX A – TARGETS WHEEL	7-1
7.1	General	7-1
7.1.1	Targets Plate construction	7-1
7.1.2	Targets wheel construction.....	7-1
7.2	Operation	7-3
7.2.1	Positioning a Target.....	7-3
7.3	Calibration	7-4
7.3.1	Managing the Axis (Targets Wheel) parameters.....	7-4
7.3.1	Calibrating the 'Home' (reference) position of the Targets Wheel.....	7-4
7.3.2	Managing a single Target information	7-4
7.3.3	Managing the Targets Plates placement.....	7-4
7.3.4	Tuning Target Relative placement	7-5
7.3.5	Testing Target absolute placement.....	7-5
8.	INDEX.....	8-1

List of Tables

Table 1-1:	ControlMaster Mechanical specification	1-3
Table 1-2:	ControlMaster Electrical specification.....	1-3
Table 1-3:	SR-800N performance (common for all types of Blackbody heads).....	1-3
Table 2-1:	ControlMaster Overview - Controls & Indicators	2-8
Table 2-2:	ControlMaster Overview - Connectors	2-9
Table 2-3:	ControlMaster Overview - RS232 Connector Pinout	2-10
Table 2-4:	ControlMaster Overview - USB Jack Pinout.....	2-10
Table 2-5:	ControlMaster Overview - LAN Jack Pinout	2-11
Table 2-6:	I/O Connector Pinout (TTL Option)	2-11
Table 2-7:	I/O Connector Pinout (Galvanic Isolation Option).....	2-12
Table 2-8:	I/O Connector Pinout (Encoder Feedback Option).....	2-12
Table 2-9:	SR-800N Cables set	2-14
Table 4-1:	I/O Control Screen - Output activation meaning	4-37
Table 4-2:	I/O Control Screen - Input indication meaning.....	4-37
Table 4-3:	'Info' screen- Fields list	4-38
Table 4-4:	Julabo Circulation System Temperature Setting.....	4-40
Table 4-5:	Preset Configuration screen- parameters list	4-50
Table 4-6:	Axis item Configuration screen- parameters list	4-58
Table 5-1:	General Failures	5-65
Table 5-2:	Blackbody Failures messages	5-67
Table 5-3:	Failures screen	5-68

List of Figures

Figure 1-1:	SR-800N Extended Area Blackbodies.....	1-1
Figure 2-1:	SR-800N Block Diagram	2-1
Figure 2-2:	SR-800N Main Components.....	2-2
Figure 2-3:	Blackbody Head Structure	2-4
Figure 2-4:	LT Blackbody Head Structure	2-6
Figure 2-5:	Julabo Circulation System	2-7
Figure 2-6:	ControlMaster Overview - Controls & Indicators	2-8
Figure 2-7:	ControlMaster Overview - Connectors	2-9
Figure 3-1:	Installation - 19" Rack Configuration.....	3-17
Figure 3-2:	Installation – LT Blackbody Head Protective cover.....	3-19
Figure 3-3:	Installation – LT Blackbody Head Pipes clamps removal.....	3-20
Figure 3-4:	Installation – LT Blackbody Head Pipes assembly	3-20
Figure 3-5:	Installation – LT Blackbody Head Pipes clamps assembly	3-21
Figure 3-6:	Installation – Julabo Circulation System Pipes assembly.....	3-21

Figure 3-7: Installation – Julabo Circulation System Cables Set	3-22
Figure 3-8: Installation – Hood assembly for LT Blackbody Head	3-23
Figure 4-1: Powering-on the SR-800N – Init message	4-26
Figure 4-2: 'Unsaved Changes' Warning message	4-27
Figure 4-3: Numeric entry screen	4-29
Figure 4-4: Alphanumeric entry screen	4-30
Figure 4-5: Absolute Mode Screen	4-31
Figure 4-6: Differential Mode Screen	4-32
Figure 4-7: Preset Selection Screen	4-34
Figure 4-8: Axis Selection screen	4-35
Figure 4-9: I/O Control Screen	4-37
Figure 4-10: 'Info' screen	4-38
Figure 4-11: Main Menu screen	4-41
Figure 4-12: Blackbody Configuration screen	4-42
Figure 4-13: Set Point Progress setting screen	4-43
Figure 4-14: Advanced Configuration screen	4-44
Figure 4-15: Temperature precision setting screen	4-45
Figure 4-16: Advanced Configuration screen - Power-Up Preset setting screen	4-45
Figure 4-17: Restore the factory settings warning message	4-47
Figure 4-20: Preset Configuration screen	4-50
Figure 4-21: Preset 'Mode' configuration Screen	4-51
Figure 4-22: Axis Setting screen	4-52
Figure 4-23: Axis Placement screen	4-53
Figure 4-24: Targets Plate Selection screen	4-54
Figure 4-25: Axis Configuration screen	4-55
Figure 4-26: Step Size Selection screen	4-57
Figure 4-27: Axis item Configuration screen	4-58
Figure 4-28: Item 'Type' Selection Screen – Type of targets	4-60
Figure 4-29: Target Tuning screen	4-62
Figure 4-30: Direct Operation screen	4-63
Figure 5-1: Failures screen	5-66
Figure 5-2: 4 ÷ 20" Blackbody Heads Emitter Sensor Cover (1/2)	5-72
Figure 5-3: 4 ÷ 20" Blackbody Heads Emitter Sensor Cover (2/2)	5-73
Figure 5-4: 4 ÷ 20" Blackbody Heads Emitter Sensor connector	5-73
Figure 5-5: 4 ÷ 20" Blackbody Heads Emitter Sensor pulling out.	5-74
Figure 5-6: 4 ÷ 20" Blackbody Heads Emitter Sensor Port Cleaning	5-74
Figure 5-7: Greasing the Sensor	5-75
Figure 5-8: 4 ÷ 20" Blackbody Heads Target Sensor connector	5-76
Figure 5-9: 2" Blackbody Head Emitter Sensor Replacement	5-77
Figure 5-10: 2" Blackbody Head Emitter Sensor pulling out	5-78

Figure 5-11: 2" Blackbody Head Target Sensor pulling out	5-79
Figure 5-12: Upgrade screen - Loading Calibration data screen (1/5)	5-80
Figure 5-13: Upgrade screen - Loading Calibration data screen (2/5)	5-81
Figure 5-14: Upgrade screen - Loading Calibration data screen (3/5)	5-81
Figure 5-15: Upgrade screen - Loading Calibration data screen (4/5)	5-82
Figure 5-16: Upgrade screen - Loading Calibration data screen (5/5)	5-82
Figure 5-17: Upgrade screen - Upgrade Software screen (1/4)	5-83
Figure 5-18: Upgrade screen - Upgrade Software screen (2/4)	5-84
Figure 5-19: Upgrade screen - Upgrade Software screen Warning.....	5-84
Figure 5-20: Upgrade screen - Upgrade Software screen Canceled	5-84
Figure 5-21: Upgrade screen - Upgrade Software screen (3/4)	5-85
Figure 5-22: Upgrade screen - Upgrade Software screen (4/4)	5-85
Figure 6-1: Setting Up the Host Computer for Direct Link.....	6-1
Figure 6-2: Setting Up the Internet Protocol (TCP/IP).....	6-2
Figure 7-1: Targets Plate construction (for reference).....	7-1
Figure 7-2: Targets wheel construction(for reference)	7-2

List of Acronyms and Abbreviations

AC:	Alternate Current
ATE:	Automated Test Equipment
BB:	Blackbody
BIT:	Built In Test
CCW:	Counterclockwise
CW:	Clockwise
DC:	Direct Current
EFL:	Effective Focal Length
HCC	Head Control Card
I/O:	Input Output
IP:	Internet Protocol
IR	Infra-Red
LAN:	Local Array Network
LCD:	Liquid Crystal Display
MCC:	Motion Control Card
RTD	resistance temperature detectors
NIST:	National Institute of Standards and Technologies
PA:	Power Amplifier
PS:	Power Supply
SBC:	Single Board Computer
TSC:	Temperature Sense Card
TTL:	Transistor Transistor Logic
UI:	Human Interface

Applicable Documents

N/A

Safety Instructions Summary

Safety Conventions

In this manual, safety information is presented as warnings, cautions, and notes.

WARNING

An operating procedure, practice, and so forth, which if not correctly followed, could result in personal injury, or loss of life.

CAUTION

An operating procedure, practice, and so forth, which if not strictly observed, could result in damage to, or destruction of equipment.

NOTE

An operating procedure, practices, and so forth, which is essential to highlight.

General Safety Instructions

WARNING

Handle this product only as instructed in this manual. Do not attempt to operate or maintain this product in a manner not specifically stated in this manual. Failure to comply could result in injury to personnel and/or damage to equipment.



WARNING

The fuse is powered even when the SR-800N is OFF. Before replacing the fuse, power-off the SR-800N remove the power cord from mains. Failure to comply may result in injury to personnel and/or damage to equipment.



HOT SURFACE

Do not attempt to touch the Blackbody surfaces during operation, it may be hot. If necessary, use the Blackbody carrying handle. Failure to comply may result in injury to personnel.



CAUTION

Power-off the SR-800N before disconnecting or replacing any of the SR-800N cables and harnesses. Failure to comply may result in damage to equipment.

Always observe standard safety precautions during the installation, operation, and maintenance of this product.

1. Introduction & General information

This chapter serves as an introduction to the SR-800N Extended Area Blackbody System, describing its main features, technical data and specifications.

1.1 SR-800N Extended Area Blackbody System Overview

SR-800N Extended Area Blackbody System consists of a high emissivity and uniformity Blackbody with its controller (the ControlMaster).

The ControlMaster measures the temperature of the Blackbody with high precision to achieve high temperature accuracy and stability.

The Blackbodies are available in various emitter sizes and temperature ranges.

The Blackbodies are equipped with removable temperature sensor (RTD) which enables fast and easy calibration of the Blackbody by replacing the sensor with new calibrated one and updating the calibration parameters, which are stored in the Blackbody head.

The customer may add accessories such as target wheel, Filter wheel etc..

The Blackbody can be operated in absolute mode – according to the temperature of the emitter, or in differential mode – the emitter temperature is relative to temperature of the target in front of it.

SR-800N Extended Area Blackbody System can be operated as a stand-alone system with local operation using the front panel of the ControlMaster, or may be integrated as part of an Electro-optics Test System, with remote operation via the communication ports of the ControlMaster.

Figure 1-1: SR-800N Extended Area Blackbodies



1.2 SR-800N Main Features

The main features of the SR-800N Extended Area Blackbody system are:

- Superior resolution.
- High uniformity.
- Wide range of operation temperature.
- Easy periodic calibration using the removable temperature sensor.
- Built-in tests to shorten maintenance time.
- Flexible ControlMaster, enables to control various types of Blackbody heads.
- User configurable resolution and stability window.
- Ethernet and RS232 communication ports (Optional GPIB on request).
- Remote operation via one of the communication ports.
- Color LCD with Touch screen for user interface.
- Controlling up to four motorized axes (optional as Filter wheel, Target wheel or sources stage)
- General purpose I/O ports (optional as TTL, dry switch or optically isolated).
- Compact and portable ControlMaster enclosure, easily installed also in 19" rack.
- Certificate of temperature calibration traceable to NIST.

1.3 General Specifications

1.3.1 ControlMaster Mechanical specification

Table 1-1: ControlMaster Mechanical specification

Operating Temperature:	0°C to 50°C
Storage Temperature:	-20°C to 70°C
Relative Humidity:	95% non condensing
Dimensions:	342x310x133 mm (3U, 63HP)
Weight:	10 Kg

1.3.2 ControlMaster Electrical specification

Table 1-2: ControlMaster Electrical specification

Operating Voltage:	110/230VAC, 50/60 Hz
Maximal current (fuse rating)	10A

1.3.3 SR-800N performance (common for all types of Blackbody heads).

Table 1-3: SR-800N performance (common for all types of Blackbody heads).

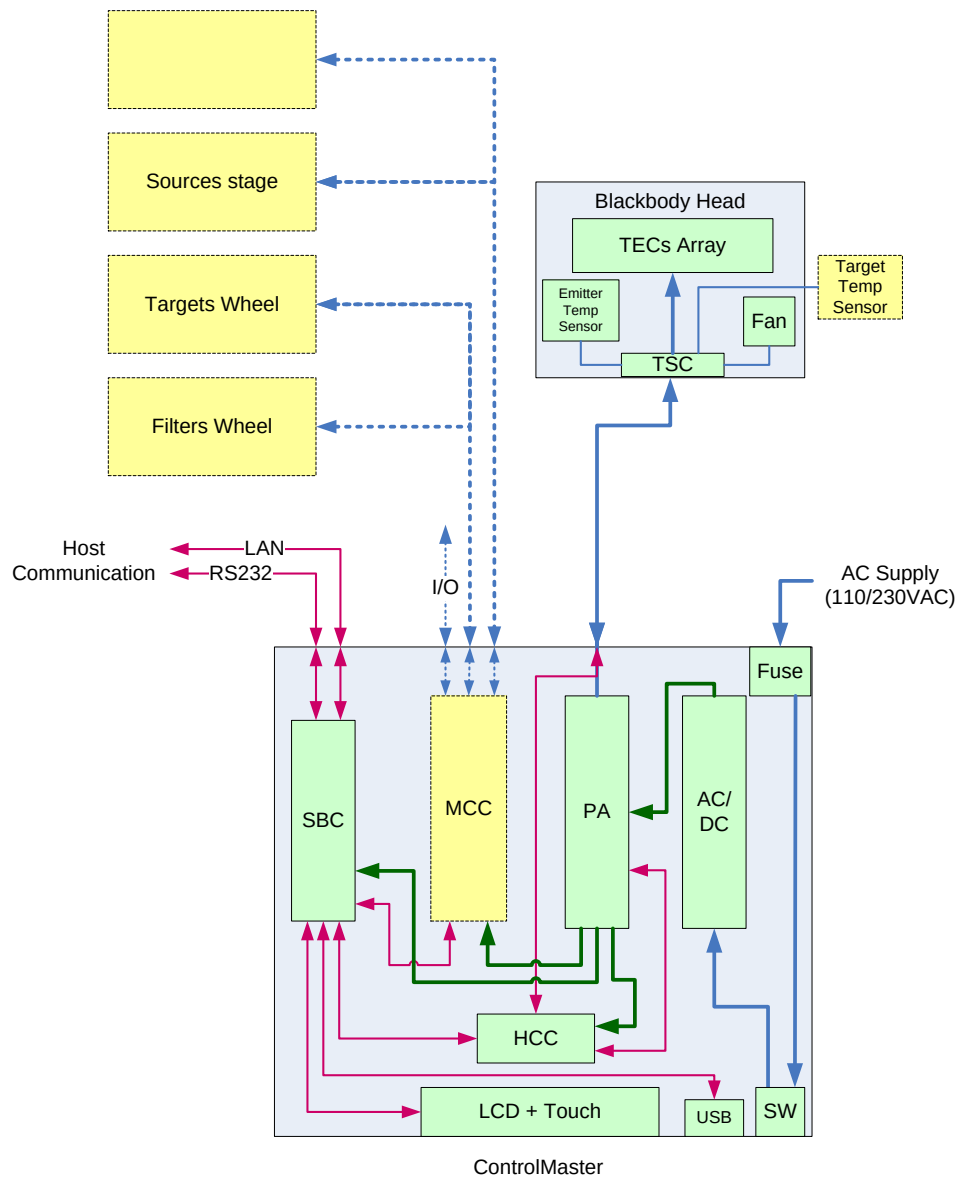
Temperature stability	
Emitter Temperature < 100 °C	±0.01°C
Emitter Temperature ≥ 100 °C	±0.01°C
Temperature reading and setting resolution	0.001°C

2. Description

This chapter describes the structure and components of the SR-800N.

2.1 SR-800N Block Diagram

Figure 2-1: SR-800N Block Diagram



2.2 SR-800N Main Components

The following describes the main components of the SR-800N System:

Figure 2-2: SR-800N Main Components



- **Blackbody Head:**
An IR radiation source based on a temperature stabilized surface (emitter) with high emissivity.
Various types of Blackbody Heads are available with different emitter size and temperature ranges (for further information regarding the types and their specifications refer to the SR-800N data sheets).
- **SR-800N ControlMaster:**
Contains the power drivers and temperature control algorithm of the Blackbody head.
Serves as electrical interface of the SR-800N System to the main supply voltage, external devices and remote operation.
LCD with touch screen in its front panel provides local User Interface.
- **Host Computer:**
Used for remotely control and monitor the SR-800N system, using one of the communication ports of the ControlMaster.
- **Motorized Axes:**
Optional in the SR-800N System.
A Motorized Axis can be used for positioning a selected target or a filter, which are installed in a dedicated wheel (i.e. Targets Wheel and Filters Wheel), in front of the emitter of the Blackbody Head.
Other usage of a Motorized Axis is positioning a stage which holds a Blackbody Head or other type of illumination source (i.e. Source stage).
The position of each Motorized Axis is controlled using stepper motors, with sensor at a reference position (i.e. Home position).

The Motorized Axes can be operated locally, from the ControlMaster, or remotely using the communication ports of the ControlMaster.

➤ **I/Os:**

Optional in the SR-800N System.

The default implementation is 6 TTL type I/Os, 2 of them configured as inputs and 4 as outputs.

There is an option for different arrangement of these I/Os, for example: 8 inputs, 8 outputs, 4 input with 4 outputs etc.

There is an option for special I/Os types when there is a need for connecting other system to the SR-800N while maintain galvanic isolation between them.

Other optional usages are to provide 'Dry switch' input (simulating a switch) to external device or system and operating low power loads (lamps, LEDs, relays etc.)

➤ **Encoder feedback:**

Optional in the SR-800N System.

This option can be used to achieve high positioning accuracy and repeatability when using a motorized axis (Targets Wheel for example).

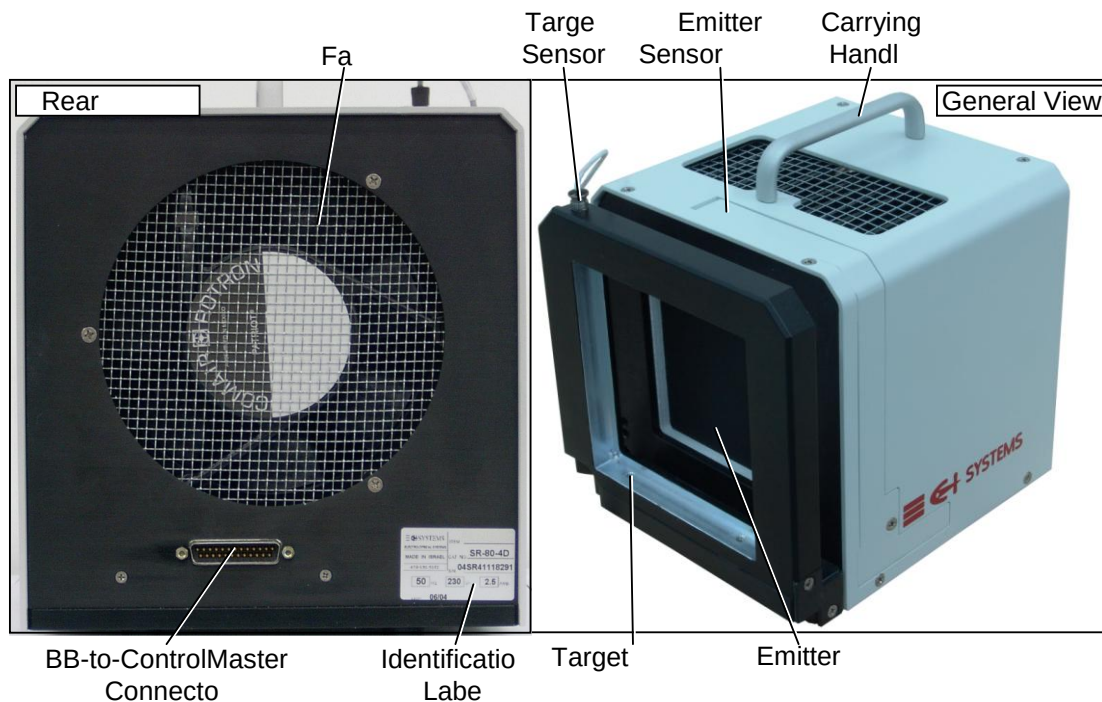
2.3 Blackbody Head Overview

2.3.1 Blackbody Head Structure

CAUTION

Damaged or scratched painted surface degrades its emissivity. Always handle the Blackbody Head and Targets with care and avoid touching the painted surfaces with bare hands.

Figure 2-3: Blackbody Head Structure



The Blackbody Head (BB) consists of the following components:

- Emitter, painted with high emissivity paint.
- RTD Temperature Sensor for the emitter (under service cover).
- Target holder (optional).
- RTD Temperature Sensor for the target (optional).
- A Heat sink with a fan (or fans) is used for cooling the emitter and stabilizes it at low temperature levels.
- Interface connector.

- Temperature Sensor Card (TSC, installed inside the head) used for:
 - Reading the emitter and target temperature.
 - Storing the temperature sensors calibration data.
 - Storing the Blackbody Head identification information (Model, S/N, etc.).
- The temperature of the emitter is stabilized by a set of Thermoelectric Cooling (TEC) devices which control the heat flow (quantity and direction) between the emitter and the heat sink.
The current through the TECs is controlled by the Power Amplifier in the ControlMaster.

2.3.2 Absolute Mode Blackbody Head

In the Absolute Mode the emitter of the Blackbody Head is thermoelectrically heated or cooled to absolute temperature.

2.3.3 Differential Mode Blackbody Head

In the differential Mode the Blackbody Head is equipped with additional target plate, clamped to the target holder, in front of the emitter surface.

The ControlMaster stabilize the temperature difference between the emitter and the target.

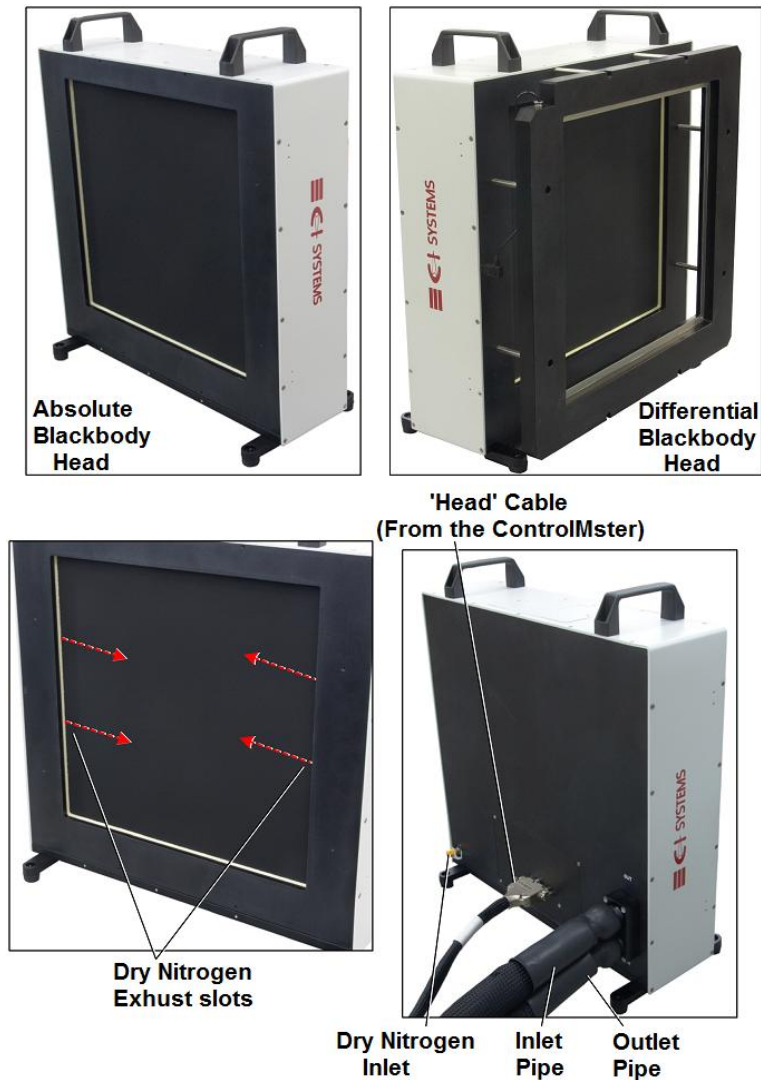
A Blackbody Head with a target and a target temperature sensor can be operated in 'Differential' mode or in 'Absolute' mode, selectable by the operator.

NOTE

The temperature range in the 'Differential' mode is
limited to the range of the 'Absolute' mode.

2.3.4 LT Blackbody Head Structure

Figure 2-4: LT Blackbody Head Structure



The LT Blackbody Head is similar to the standard Blackbody Head with the following changes:

- The heat sink and fan (or fans) are replaced with heat exchanger, through which the cooling liquid is circulated.
- Inlet and Outlet pipes are used to connect the Julabo Circulation System to the heat exchanger in the Blackbody Head.
- Dry nitrogen flow can be supplied using the inlet connector. Dedicated exhaust slots, located around the Emitter, are used to spread it to prevent condensation and icing at temperature below 5°C.

NOTE

The Temperature uniformity of the Emitter may degrade when using the Dry Nitrogen.

2.3.4.1 Julabo Circulation System

Figure 2-5: Julabo Circulation System



The Julabo Circulation System is used to stabilize the temperature of the cooling fluid and to circulate it through the LT Blackbody Head.

The Julabo Circulation System has the following main components:

- Cooling liquid Tank.
- Circulation pump.
- Controller for liquid temperature and liquid flow control.
- For more information refer to the '*Julabo Circulation system operation manual*'.

2.4 ControlMaster Overview

2.4.1 ControlMaster Components

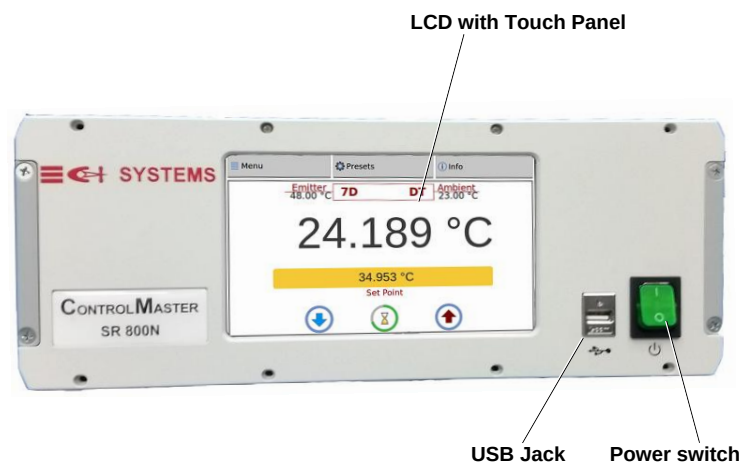
- See ControlMaster block diagram ([Figure 2-1](#)).
- LCD with Touch panel - for local User Interface (UI)
- Single Board Computer (SBC) – for managing the SR-800N System, operating the communication channels and the UI.
- Head Control Card (HCC) – holds the Blackbody temperature control algorithm.
- AC/DC – main power supply of the SR-800N System.
- Power Amplifier (PA) – supplies internal voltages (24V and 12V) and drives the controlled current to the TECs in the Blackbody head.
- Optional: Motors Control Card (MCC) – drives up to 4 axes and special I/Os.

2.4.2 Controls & Indicators

Table 2-1: ControlMaster Overview - Controls & Indicators

#	Name	Description
1.	Power switch	Two-position green colored switch used to turn ON/OFF the power supply to the SR-800N.
2.	LCD with Touch Panel	LCD with touch panel for local user interface
3.	USB jack	For connecting mass storage device for sensors calibration and S/W upgrade purpose.

Figure 2-6: ControlMaster Overview - Controls & Indicators

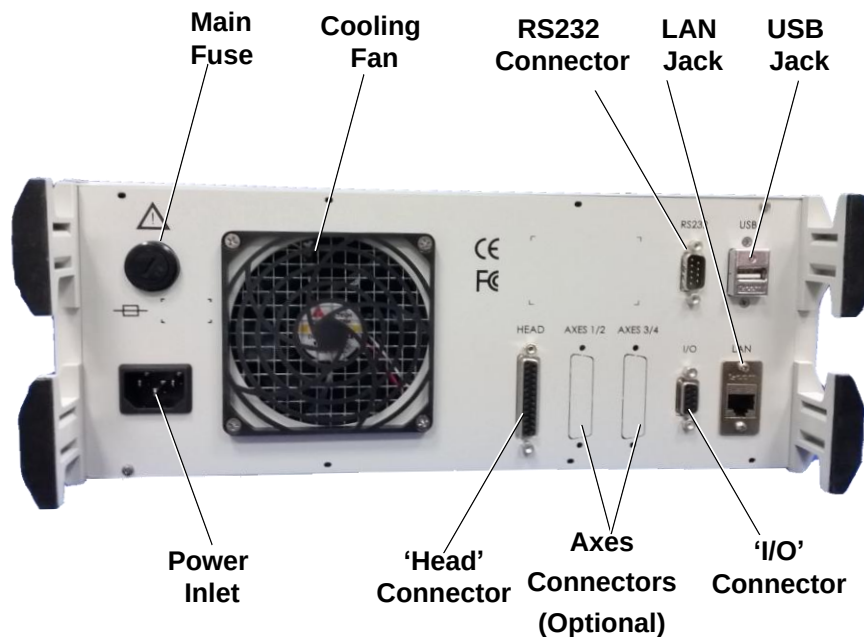


2.4.3 Connectors

Table 2-2: ControlMaster Overview - Connectors

#	Name	Description
1.	Power in jack	Standard IEC power inlet for AC power (110 or 230VAC).
2.	Main Fuse	Main fuse for the AC supply power
3.	Head	For connecting the Blackbody head to the ControlMaster
4.	RS232	Male 9 pin D-type for RS232 communication.
5.	Rear USB	Type A USB jack, for connecting USB devices to the ControlMasterControlMaster.
6.	LAN	RJ45 for Ethernet communication.
7.	I/O	Female 9 pin D-type for general purpose I/O signals. Optional special I/O types.
8.	AXES 1/2	<u>Optional</u> – for connecting 2 motorized Axes (marked 'Axis 1' and 'Axis 2') to the ControlMasterControlMaster. This slot in the rear panel ControlMasteris blocked when not used.
8.	AXES 3/4	<u>Optional</u> – for connecting 2 motorized Axes (marked 'Axis 3' and 'Axis 4') to the ControlMasterControlMaster. This slot in the rear panel ControlMasteris blocked when not used.

Figure 2-7: ControlMaster Overview - Connectors



2.4.4 RS232 Connector Pinout

Table 2-3 lists the RS232 Connector Pinout per EIA/TIA 232:

Table 2-3: ControlMaster Overview - RS232 Connector Pinout

Pin #	Signal Name	Function	Voltage (V)	Current (mA)
1	----			
2	RXD	Received Data (in)	± 25	± 15
3	TXD	Transmitted Data (out)	± 25	± 15
4	----			
5	GND	Ground		
6	----			
7	----			
8	----			
9	----			

2.4.5 USB Jack Pinout

Table 2-4 lists the USB Jack Pinout (Front and Rear):

Connector Type – USB A, female, the ControlMaster is a Host.

Table 2-4: ControlMaster Overview - USB Jack Pinout

Pin #	Signal Name	Function	Max Voltage (V)	Max Current (mA)
1	V_BUS	5V for device	5	500 (max)
2	D+	Data +	5	----
3	D-	Data -	5	----
4	GND	Ground	5	----

2.4.6 LAN Jack Pinout

Table 2-5 lists the LAN Jack Pinout per EIA/TIA 568A/B
Connector Type – RJ45

Table 2-5: ControlMaster Overview - LAN Jack Pinout

Pin #	Signal Name	Function	Note
1	TXP_D1	Transmitted Data+	
2	TXN_D1	Transmitted Data -	
3	RXP_D2	Received Data +	
4	BIP_D3	Bi-Directional+	
5	BIN_D3	Bi-Directional-	
6	RXN_D2	Received Data -	
7	BIP_D3	Bi-Directional+	
8	BIN_D3	Bi-Directional-	

2.4.7 I/O Connector Pinout (TTL Option)

Table 2-6 lists the I/O Connector Pinout (TTL Option):
Connector Type – D-sub 9 pin (Female)

Table 2-6: I/O Connector Pinout (TTL Option)

Pin #	Signal Name	Function	Voltage (V)	Current (mA)
1	OUT1	Output	TTL (5)	5 (max)
2	OUT3	Output	TTL (5)	5 (max)
3	GND	Ground	----	----
4	IN1	Input	TTL (5)	
5	IN2	Input	TTL (5)	IN1
6	OUT2	Output	TTL (5)	5 (max)
7	OUT4	Output	TTL (5)	5 (max)
8	OUT5	Output	TTL (5)	5 (max)
9	GND	Ground	----	----

2.4.8 I/O Connector Pinout (Galvanic Isolation Option)

Table 2-7 lists the I/O Connector Pinout (Galvanic Isolation Option):
Connector Type – D-sub 9 pin (Female)

Table 2-7: I/O Connector Pinout (Galvanic Isolation Option)

Pin #	Signal Name	Function	Voltage (V)	Current (mA)
1	SW1_NO	NO Dry switch Output	24	1
2	SW1_NO			
3	Out3_OC	Opencollector output	24	0.2
4	IN1	Optocoupler Input	5	10
5	IN2	Optocoupler Input	5	10
1	SW2_NO	NO Dry switch Output	24	1
2	SW2_NO			
8	Out4_OC	Opencollector output	24	0.2
9	GND_REF	Opencollector GND REF	----	----

2.4.9 I/O Connector Pinout (Encoder Feedback Option)

Table 2-8 lists the I/O Connector Pinout (Encoder Feedback Option):
Connector Type – D-sub High Density 15 pin (Female)

Table 2-8: I/O Connector Pinout (Encoder Feedback Option)

Pin #	Signal Name	Function	Voltage (V)	Current (mA)
1	ENC P	Option for limit switch input	5	
2	ENC q	Option for limit switch input	5	
3	Not used			
4	ENC I_L	Index Diff. Channel	RS422 (5)	
5	ENC B_L	B Diff. Channel	RS422 (5)	
6	ENC A_L	A Diff. Channel	RS422 (5)	
7	P5V	5V supply	5	100
8	Not used			

Pin #	Signal Name	Function	Voltage (V)	Current (mA)
9	GND	Ground	----	----
10	Not used			
11	Not used			
12	ENC I_H	Index Diff. Channel	RS422 (5)	
13	ENC B_H	B Diff. Channel	RS422 (5)	
14	ENC A_H	A Diff. Channel	RS422 (5)	
15	Not used			

2.5 SR-800N Cables set

Table 2-9 lists the SR-800N cables and harnesses.

Table 2-9: SR-800N Cables set

	Name	Description	Note
1.	Power Cord	AC Supply to the SR-800N, from the mains.	Mates with IEC C14 jack
2.	Head Harness	Dedicated cable to connect the Blackbody head with the ControlMaster.	
3.	AXES 1/2	Dedicated cable to connect 2 motorized devices (Filters/Targets Wheel with the ControlMaster.	Optional
4.	AXES 3/4	Dedicated cable to connect 2 motorized devices (Filters/Targets Wheel with the ControlMaster.	Optional
5.	I/O	Dedicated cable to the I/O connector (9 pin female D-Type)	Optional
6.	Encoder	Dedicated cable to the Encoder connector (15 pin female HD D-Type)	Optional
7.	LAN cable	Standard LAN cable (at least Cat5).	Not part of the SR-800N deliverables
8.	RS232 Comm.	Mate with the 9 pin male D-Type connector.	

NOTE

For maximal lengths – contact CI Systems.

2.6 SR-800N Main Options

2.6.1 Motorized Axes

This option is used to operate up to 4 different axes - each may activate a targets wheel or filters wheel or a stage.

2.6.2 Encoder feedback

This option, when used with the motorized axes option, enables high accuracy and repeatability of the positioning of the motorized axis (for example Targets wheel) using a feedback from incremental encoder.

2.6.3 Special I/Os

This option includes several types of I/Os, enabling to engage the SR-800N to a different system, while maintaining galvanic isolation, or to use the ControlMaster to operate low power loads (lamps, LEDs, relays etc.).

2.6.4 Extended Temperature Range

This option (marked ET) is used to operate the Blackbody Head in wider range, while maintain the standard performance.

2.6.5 Low Temperature Blackbody Heads

This option (marked LT) is used to operate the Blackbody Head at very low temperature. The fan and heat-sink of the Blackbody Head are replaced with external cooler and liquid circulator. This option requires a hood with nitrogen/dry air stream to prevent icing of the emitter.

2.6.6 Wide Temperature Range

This option (marked WTR) is used to operate the Blackbody Head in its widest range, mainly well below the 0°C (absolute), without the need of external cooler. This option requires a hood with nitrogen/dry air stream to prevent icing of the emitter

2.6.7 High Emissivity Emitter

In this option the emitter of the Blackbody Head is replaced with special one, with very high emissivity.

2.6.8 Emissivity Compensation

The compensation is done by changing the actual temperature of the emitter so that its radiated spectra, at the specified wavelength range, matches the Plank function at the set point temperature.

2.6.9 High Uniformity

This is a calibration option, which includes additional uniformity test and calibration.

3. Handling & Installation

This chapter explains how to transport, store, unpack, and install the SR-800N.

3.1 Transportation

The SR-800N can be transported by land, sea, or air while installed in its original package.

3.2 Storage

While installed in its original package, the SR-800N can be stored in the following conditions:

- Temperature range: -20°C to +70°C
- Relative humidity: 5% - 95% (non condensing)

3.3 Installation

CAUTION

Screws shall be tightened using hand-torque.

Failure to comply may result in damage
to equipment.

3.3.1 Unpacking

- Verify that the number of packages in the delivery matches the number of packages stated in the attached packing list.
- Visually inspect all packages for traces of excessive moisture or external damage.
- Carefully open each package, observing handling labels ("this side up", "open this end", and the like).

3.3.2 Installation Configurations

The SR-800N can be installed in the following configurations:

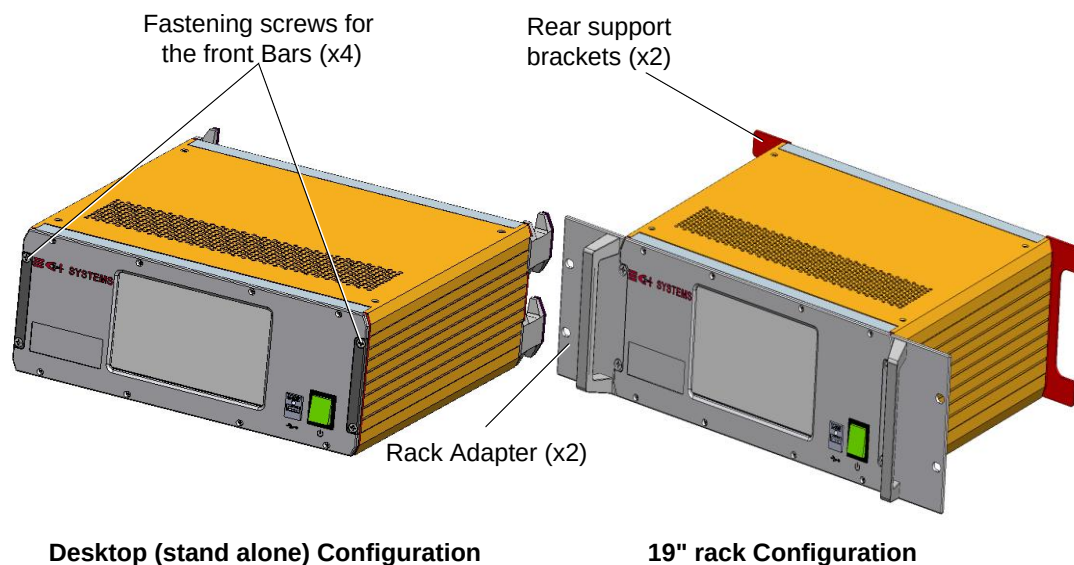
3.3.2.1 Desktop (stand alone) Configuration

- Unpack the SR-800N (see 3.3.1).
- Place the SR-800N ControlMaster on the Bench, unfold its front feet.
- Place the SR-800N Blackbody head on the Bench.
- Connect the cables set to the rear panel (see Figure 2-7):
 - Connect the power cord between the ControlMaster power in jack and Mains
 - Connect the 'HEAD' cable between the ControlMaster 'HEAD' connector and the Blackbody connector.
 - Connect all the other standard cables if applicable (such as LAN, RS232).

3.3.2.2 19" Rack Configuration

- Unpack the SR-800N (see Section 3.3.1).
- Remove the bars on each side of the ControlMaster 's front panel by releasing their fastening screws.
- Use these screws to place and secure the Rack Adapters on both sides.

Figure 3-1: Installation - 19" Rack Configuration



NOTE

It is recommended to place the ControlMaster on rails inside the rack. For this, replace the rear feet (4 units) with the rear support brackets (2 units) (see [Figure 3-1](#)).

- Install the SR-800N ControlMaster in the 19" rack.
- Place the SR-800N Blackbody head on the Bench.
- Connect the cables set to the rear panel (see [Figure 2-7](#)):
 - Connect the power cord between the ControlMaster power in jack and Mains
 - Connect the 'HEAD' cable between the ControlMaster 'HEAD' connector and the Blackbody connector
 - Connect all the other standard cables if applicable (such as LAN, RS232).

3.3.3 Low temperature (LT) Blackbody Head Installation

3.3.3.1 Safety Precautions

- Read all safety instructions stated in the *Julabo Circulation System Operating Manual*. Follow the hazards in this manual to avoid possible injuries to personnel and damage to equipment.
- Verify correct level of cooling fluid level before starting the circulator, and after short operation time (it may drop as the fluid fills the circulation piping). Refill to the required level if needed.

3.3.3.2 Unpacking

- Unpack the Julabo Circulation System and place it on site (refer to the *Julabo Circulation System Operating Manual*).
- Unpack the Blackbody Head and place it near the Julabo Circulation System.

NOTE

The maximal distance between the units is 4 m.

- If the Blackbody Head is covered: release four captive screws that fasten the Blackbody Protective Cover and remove it.

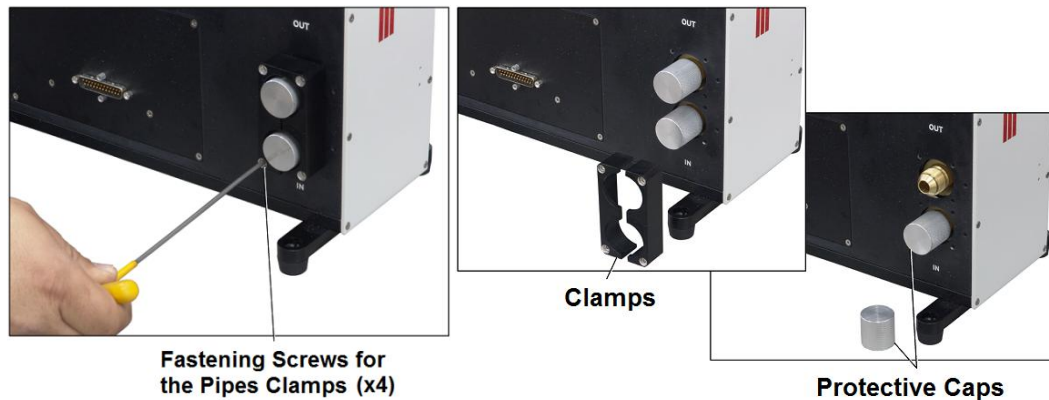
Figure 3-2: Installation – LT Blackbody Head Protective cover



3.3.3.3 LT Blackbody Head - Connecting cooling liquid and Dry Nitrogen pipes

- Release the fastening screws of clamps of the pipes at rear of the Blackbody Head and remove the clamps.
- Remove the caps of the pipe inlets and store them.

Figure 3-3: Installation – LT Blackbody Head Pipes clamps removal



- Connect and fasten the pipes using a 22 mm open-wrench

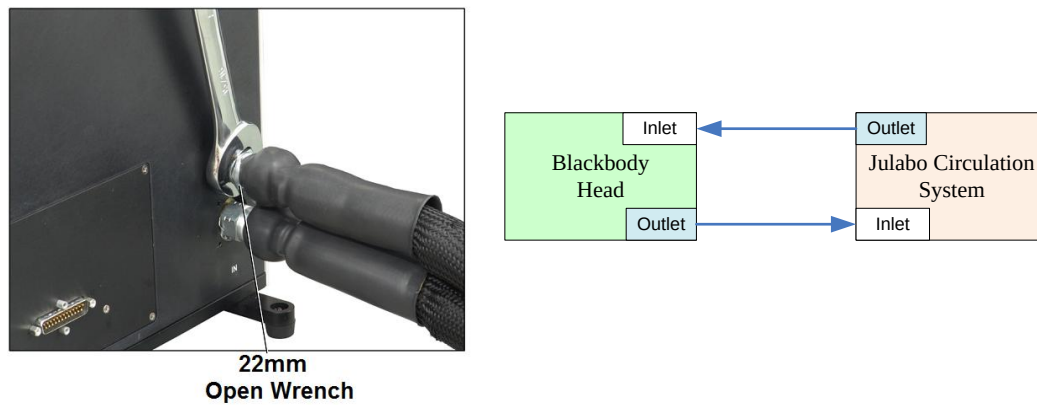
CAUTION

Verify proper pipes connection:

The Blackbody Head inlet is connected to the Julabo Circulation System outlet,

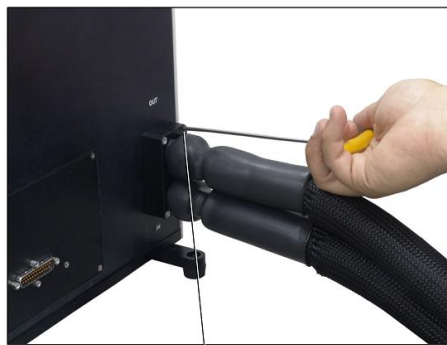
The Blackbody Head outlet is connected to Julabo Circulation System inlet.

Figure 3-4: Installation – LT Blackbody Head Pipes assembly



- Reassemble and fasten the pipe clamps.

Figure 3-5: Installation – LT Blackbody Head Pipes clamps assembly



**Fastening Screws for
the pipe clamps (x4)**

- If needed, connect a Dry Nitrogen Source to the Blackbody Head Dry Nitrogen Inlet.

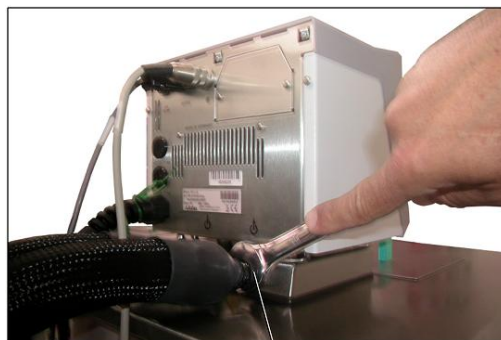
NOTE

The recommended maximal Dry Nitrogen pressure is **1bar.**

3.3.3.4 Julabo Circulation System - Connecting cooling liquid pipes

- Connect and fasten the pipes using a 19 mm open-wrench.

Figure 3-6: Installation – Julabo Circulation System Pipes assembly



**19mm
Open Wrench**

CAUTION

Verify proper pipes connection:

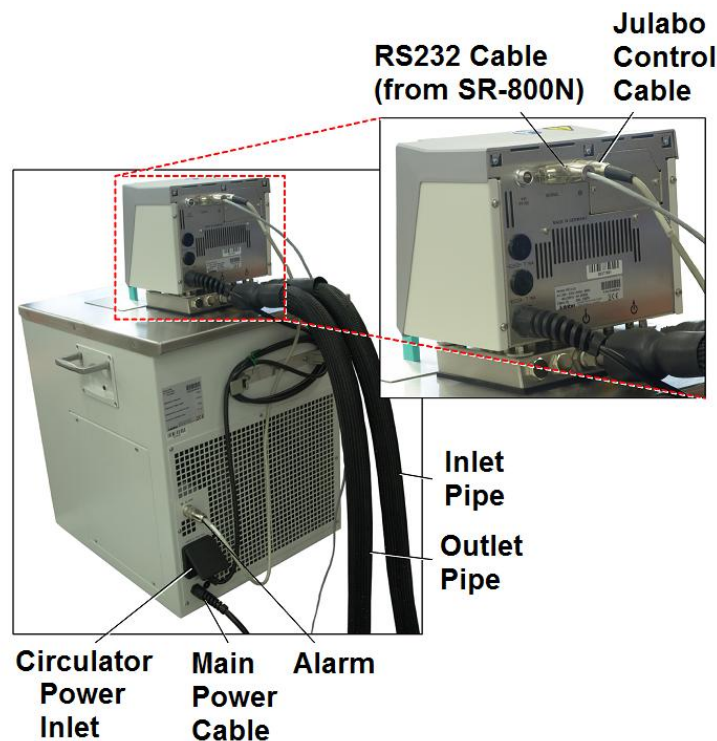
Blackbody Head inlet to Julabo Circulation System outlet,

Blackbody Head outlet to Julabo Circulation System inlet.

3.3.3.5 Cables Set Connection

- Connect the SR-800N cables set to the rear panel (see [Figure 2-7](#)):
 - Connect the power cord between the ControlMaster power in jack and Mains.
 - Connect the 'HEAD' cable between the ControlMaster 'HEAD' connector and the Blackbody connector.
 - Connect the 'RS-232' cable between the ControlMaster 'RS-232' connector and the 'Serial' connector of the Julabo Circulation System.
 - Connect all the other standard cables if applicable (such as LAN).
- Connect the Julabo Circulation System cables set:
 - Connect the Control Cable between the Controller and the 'ALARM' jack on the circulator.
 - Connect the Power Cord to the Circulator Power inlet.
 - Connect the Main Power Cable to the Mains (directly or via a 100VAC transformer).

Figure 3-7: Installation – Julabo Circulation System Cables Set



3.3.3.6 Low Temperature Hood

- Attach the Hood to the LT Blackbody Head, use its latch to secure it.
- Connect the Dry Nitrogen Source to the Dry Nitrogen Inlet of the Hood

NOTE

The recommended maximal Dry Nitrogen pressure is **1bar**.

The Dry Nitrogen Connection to the hood replaces the connection of the one of the LT Blackbody Head
(see [Section 3.3.3.3](#))

Figure 3-8: Installation – Hood assembly for LT Blackbody Head



3.4 Removal

3.4.1 Desktop (stand alone) Configuration

- Disconnect the cables set from the rear panel (see [Figure 2-7](#)):
 - Disconnect the power cord between the ControlMaster power in jack and Mains
 - Disconnect the 'HEAD' cable between the ControlMaster 'HEAD' connector and the Blackbody connector
 - Disconnect all the other standard cables if applicable (such as LAN, RS232).
- Remove the SR-800N Blackbody head from the Bench.
- Remove the SR-800N ControlMaster from the Bench.
- Pack the SR-800N Blackbody Head and ControlMaster.

3.4.2 19" rack Configuration

- Disconnect the cables set from the rear panel (see [Figure 2-7](#)):
 - Disconnect the power cord between the ControlMaster power in jack and Mains.
 - Disconnect the 'HEAD' cable between the ControlMaster 'HEAD' connector and the Blackbody connector.
 - Disconnect all the other standard cables if applicable (such as LAN, RS232).
- Remove the SR-800N Blackbody head from the Bench.
- Remove the SR-800N ControlMaster from the 19" rack.
- Remove the 19" rack adapters.
- Pack the SR-800N Blackbody Head and ControlMaster.

3.4.3 LT Blackbody Head

3.4.3.1 Draining the Fluid from the LT Blackbody Head

- Drain the fluid from the Julabo Circulation System fluid tank (refer to the *Julabo Circulation System Operating Manual*).
- Disconnect one of the pipes from the Julabo Circulation System using a 19 mm open-wrench (see [Figure 3-7](#)).
- Attach a high pressurized air hose to the released pipe.
- Blow the cooling fluid out of the pipes and the Blackbody Head into the fluid tank of the Julabo Circulation System.
- Release and remove the Pipes from the Julabo Circulation System using a 19 mm open-wrench (see [Figure 3-7](#)).

3.4.3.2 Disconnecting the pipes from the LT Blackbody Head

- Release the fastening screws of clamps of the pipes at rear of the Blackbody Head and remove the clamps (see [Figure 3-3](#)).
- Release and remove the Pipes from the Blackbody Head using a 22 mm open-wrench (see [Figure 3-4](#)).
- Re-install the caps on the pipe inlets (see [Figure 3-3](#)).
- Re-install the clamps of the pipes and fasten their screws at rear of the Blackbody Head (see [Figure 3-5](#)).

3.4.3.3 Low Temperature Hood

- Disconnect the Dry Nitrogen pipe from the Hood
- Release the latches and remove the Hood from the Blackbody Head (see [Figure 3-8](#)).

3.4.3.4 Re-Packing

- Pack the SR-800N Blackbody Head, the ControlMaster, the Julabo Circulation System and the Hood.

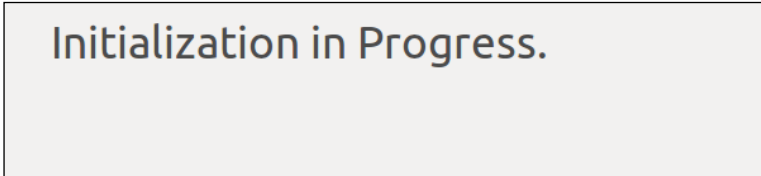
4. Operation

This chapter details the operating procedures of the SR-800N.

4.1 Powering-on the SR-800N

- Power-on the SR-800N using the Power switch on the front of the ControlMaster (see [Figure 2-6](#)).
- The power up sequence starts with collecting system configuration and calibration parameters from the Head Control Card (HCC), the Blackbody head (using communication with the TSC), the Power Amplifier (PA) and Motors Control Card (MCC), if installed.
- At this stage, the communication between the components is tested and the collected data is validated.
- If installed, the MCC initiates the axes (Targets wheel, Filters wheel or a stage) of the SR-800N to their home position.
- During the power up, the ControlMaster displays the following message:

Figure 4-1: Powering-on the SR-800N – Init message



Initialization in Progress.

- If any malfunction is detected during this procedure, the failure list notification is displayed (for more details, see [Sections 5.2](#)).
- Upon Power-up successful completion, the Main Temperature Control screen is displayed in either Absolute or Differential mode.

4.2 Turning-Off the SR-800N

- Power-off the SR-800N using the Power switch on the front of the ControlMaster (see [Figure 2-6](#)).

CAUTION

The emitter of the Blackbody head may be hot when turning off the SR-800N. Allow sufficient cooling period.

4.3 Operation Procedures

The SR-800N operation includes the following:

4.3.1 General navigation buttons

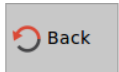
The following buttons are visible in all the configuration and setting screens.

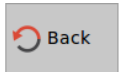
4.3.1.1 'Home' button.



Press the  button to return to the main screen (Absolute or differential, according to the system's setting)

4.3.1.2 'Back' button.



Press the  button to return to the previous screen (without saving changes).

4.3.1.3 'Save' button.



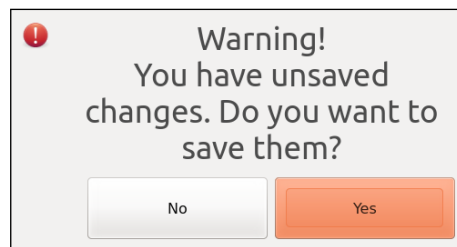
Press the  button to save the current setting (without closing the screen). This button is active only if the content of the screen was modified.

4.3.1.4 'Unsaved Changes' Warning

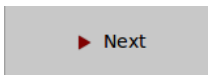
If the content of the screen was modified and not saved (save button is active), when the operator tries to close the screen by pressing the 'Home' or 'Back' buttons, a warning message pops-up.

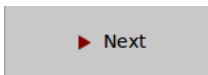
Select the 'No' button to close the screen without saving the changes or 'Yes' save them before closing the screen.

Figure 4-2: 'Unsaved Changes' Warning message



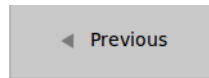
4.3.1.5 'Next' button.



Press the  button to navigate to the next configuration screen of the selected item.

Note: this button becomes inactive at the last configuration screen or item.

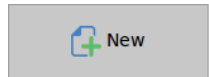
4.3.1.6 'Previous' button.



Press the button to navigate to the previous configuration screen of the selected item.

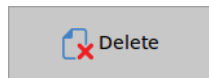
Note: this button becomes inactive at the first configuration screen or item.

4.3.1.7 'New' button.



Press the button to create a new item of the same type.

4.3.1.8 'Delete' button.



Press the button to removes the current item.

4.3.2 General Controls

The following are used to select or activate functions in the settings screens.

4.3.2.1 'Radio buttons'

'Radio buttons' are used to enable the operator to select between options:

The selected option is marked with:



All other options are marked with:



Press on the required option to select it – all other related option will become non-selected.

4.3.2.2 'ON/OFF switch'

'ON/OFF switch' are used to enable the operator to enable/activate ('ON' state) or disable/inhibit ('OFF' state) a component or option:

When the component/option is active (enabled), it is marked with:



When the component/option is inhibited (disabled), it is marked with:



Press on the switch to toggle between its states.

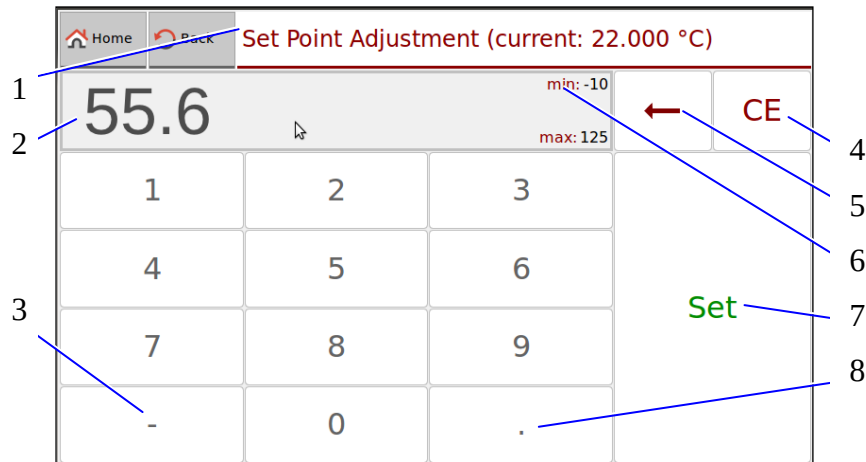
4.3.3 General Data Entry Screens

The following screens are used to enter numeric or alphanumeric values.

4.3.3.1 Numeric entry screen

Numeric entry screen is used to enter new numeric value, for example new temperature 'Set Point'.

Figure 4-3: Numeric entry screen



The items of the Numeric entry screen are:

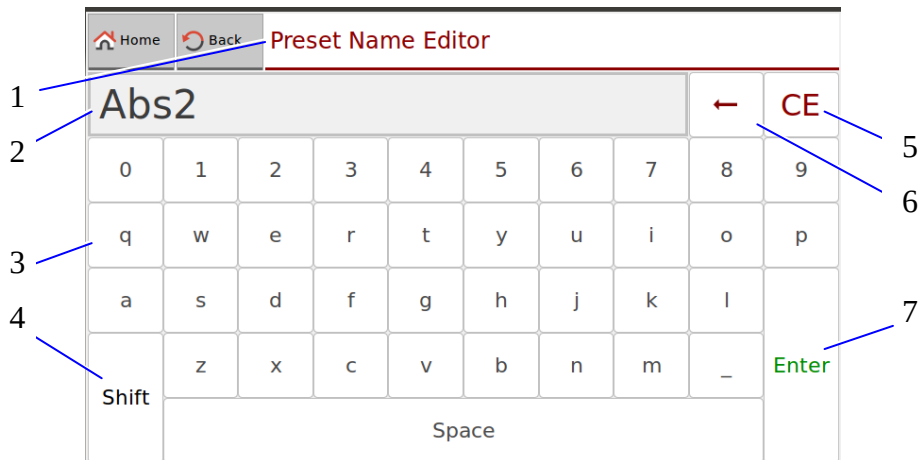
- **Item 1** - the header of the screen with the following structure:
 - The name of the function or variable that accepts the new value.
 - The current value of the function or variable (in brackets).
- **Item 2** - displays the new entered value.
- **Item 6** - indicates the upper (mark 'max') and lower (mark 'min') limits of the entered value. If the entered value exceeds these limits, it will be ignored.
- Use the displayed keyboard to enter the new value:
 - **Item 3** - the '-' key for negative numbers.
 - **Item 8** - the '.' key for fractional values.
 - **Item 4** - the 'CE' (clear entry) key to clear the entire new value.
 - **Item 5** - the '←' key (backspace) to erase the last digit.
 - **Item 7** - the 'Set' button to accept the new value and return to the previous screen.

4.3.3.2 Alphanumeric entry screen

Alphanumeric entry screen is used to enter free textual value, containing digits, upper and lower case characters.

An example of such values is name of Axis, Presets or Targets.

Figure 4-4: Alphanumeric entry screen



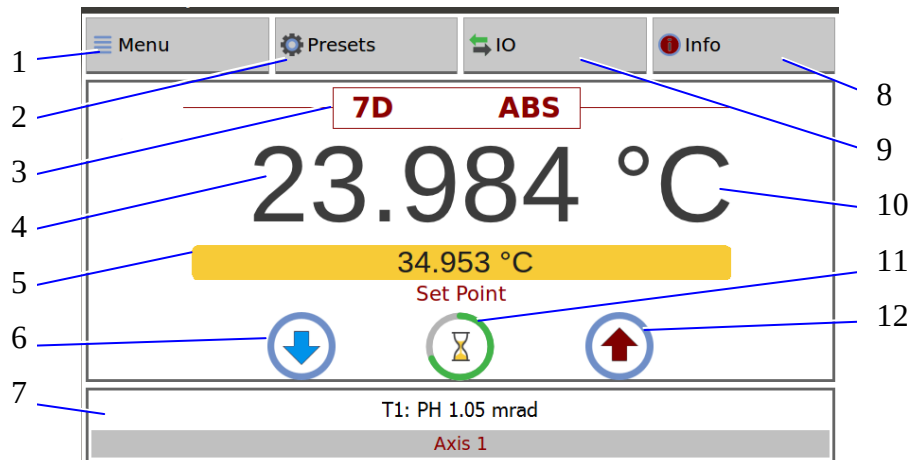
The items of the Alphanumeric entry screen are:

- **Item 1** - the header of the screen indicating the name of the function or variable which will accept the new alphanumeric value.
- **Item 2** - displays the new entered value.
- Use the displayed keyboard to enter the new value:
 - **Item 3** - a keyboard with all the available characters.
 - **Item 4** – the 'Shift' key for upper case characters.
 - **Item 5** - the 'CE' (clear entry) key to clear the entire new value.
 - **Item 6** - the '←' key (backspace) to erase the last entry.
 - **Item 7** - the '**Set**' button to accept the new value and return to the previous screen.

4.3.4 Absolute Mode Screen

Upon pre-setting mode of operation to 'Absolute' (see [Section 4.4.1.1](#)), the Absolute Mode Screen is displayed immediately after successful power-up (see [Section 4.1](#)):

Figure 4-5: Absolute Mode Screen



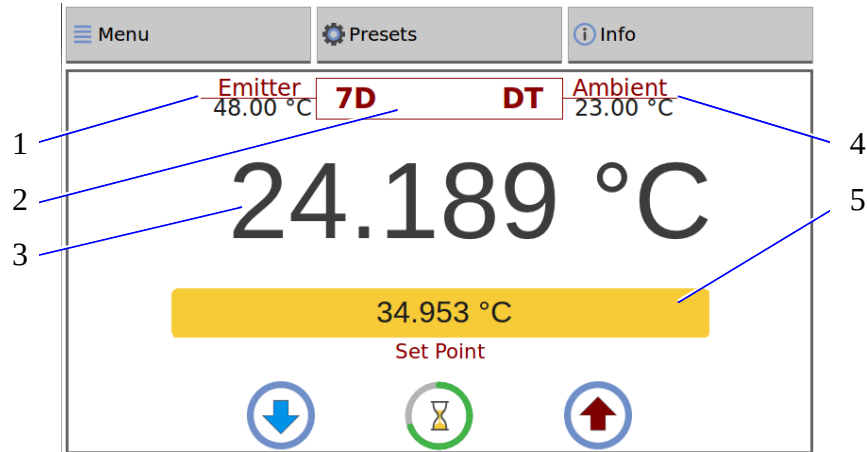
The Absolute Mode Screen has the following items:

- **Item 1** – a button for activating the main 'Menu' screen (see [Section 4.4.1](#)).
- **Item 2** – a button for activating the 'Preset' selection screen (see [Section 4.3.7](#)).
- **Item 3** – Operation Mode ('ABS' for Absolute) and Blackbody Head type ('7D' for differential 7") indicator.
- **Item 4** – Emitter's temperature readout and a button for manual temperature setting (see [Section 4.3.6.1](#)).
- **Item 5** – The current 'Set Point' for the Emitter's temperature. **Yellow background** indicates new value, while **Gray background** indicates a stable or previous value.
- **Items 6 and 12** – buttons for decreasing/increasing the set point (see [Section 4.3.6.2](#)). The rate and resolution of changing can be set by the operator (see [Sections 4.4.1.1.3 and 4.4.1.1.4](#)).
- **Item 7** – a button for activating one of the installed motorized axes (see [Section 4.3.8](#)).
- **Item 8** – a button for activating the 'Info' screen (see [Section 4.3.10](#)) with failures status indicator ( / ). The failure list can be viewed in the 'Failures' screen (see [Section 5.2.20](#)), accessed from the 'Info' screen.
- **Item 9** – a button for activating, monitoring and setting the I/Os of the ControlMaster (see [Section 4.3.9](#)).
- **Item 10** – an indicator for the temperature measuring unit (see [Section 4.4.1.1.2](#)).
- **Item 11** – an indicator for Heating/Cooling processes and temperature stabilization status (see [Section 4.3.6.3](#)).

4.3.5 Differential Mode Screen

Upon pre-setting mode of operation to 'Differential' (see [Section 4.4.1.1](#)), the Differential Mode Screen is displayed immediately after successful power-up (see [Section 4.1](#)):

Figure 4-6: Differential Mode Screen



The Differential Mode Screen has the following items, in addition to those of the Absolute Mode (see [Section 4.3.4](#)):

- **Item 1** – The Emitter's temperature readout.
- **Item 2** – Operation Mode ('DT' for Differential) and Blackbody Head type ('7D' for differential 7") indicator.
- **Item 3** – Differential temperature readout (Emitter's temperature – Ambient/Target temperature).
- **Item 4** – The Ambient temperature readout, measured at the target holder.
- **Item 5** – The Current 'Set Point' for the differential temperature.
Yellow background indicates a new value, while **Gray background** indicates a stable or previous value.

NOTE

The number of visible buttons at the top and the bottom of the screen depends on the SR-800N System configuration; only installed components are visible (see [Figure 4-5](#) and [Figure 4-6](#)).

4.3.6 Setting the Temperature

4.3.6.1 Manual Setting

- Press on the measured temperature in the main screen (Figure 4-5, item 4) to invoke the numeric entry screen (marked 'Set Point Adjustment') (see Figure 4-3).
- After setting new value, the background of the **Set Point** value (Figure 4-5, item 5) becomes yellow and the progress indication becomes  (Figure 4-5, item 11).
The size of its **green** perimeter indicates the progress of the difference between the set point and the current measured temperature.

4.3.6.2 Incremental Setting

- Press the  button to increase the set point or the  button to decrease it (Figure 4-5, items 6 and 12).
- The increment/decrement continues as long as the buttons are pressed.
- During and at the end of the increment/decrement, the background of the **Set Point** value (Figure 4-5, item 5) becomes yellow and the progress indication becomes  (Figure 4-5, item 11).
The size of its **green** perimeter indicates the progress of the difference between the set point and the current measured temperature.
- The speed of incrementing/decrementing the set point can be set using the 'Blackbody Configuration' under the main 'Menu' (see Section 4.4.1.1)

4.3.6.3 'Stable Temperature' Indication

- When the measured temperature is within the stabilization window for 30 sec, the background of the **Set Point** (Figure 4-5, item 5) value becomes light grey, and the progress indication becomes  (Figure 4-5, item 11).
- The size of the stabilization window can be set using the 'Blackbody Configuration' under the main 'Menu' (see Section 4.4.1.1)
- In addition to this visual indication, the ControlMaster uses its buzzer for audible indication when the temperature is stable. The operator can activate or inhibit the buzzer using the 'Advanced Configuration' under the main 'Menu' (see Section 4.4.1.2).

4.3.6.4 Temperature Resolution

- The resolution of the measured temperature and the set-point values is can be set using the 'Blackbody Configuration' under the main 'Menu' (see Section 4.4.1.1).

4.3.7 Preset functions

- A 'preset' is a function that enables the operator to define a specific combination of set-point and mode of operation. If Axes are installed, the operator can add selection of items from these Axes (target, filter and source).
- When a preset is selected, all its content is immediately activated.
- The operator can name each preset with meaningful value, for ease of use.

4.3.7.1 Preset Selection Screen

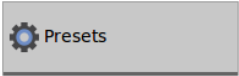
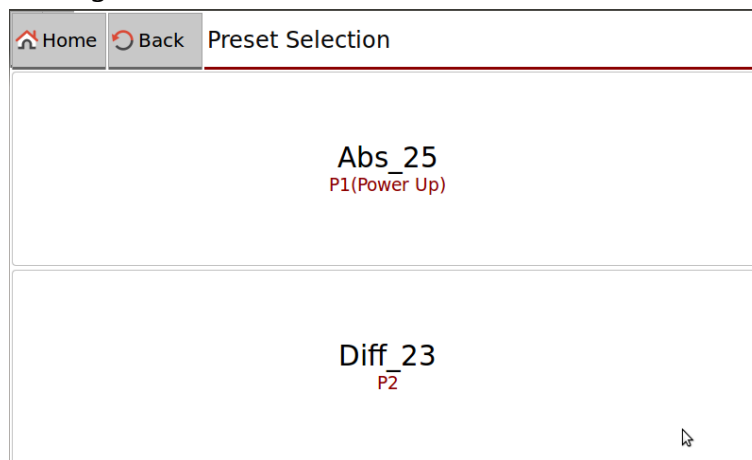
- To invoke the Preset Selection Screen press the  button (located at the upper side of the main Absolute or Differential mode screen):

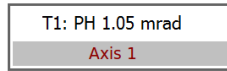
Figure 4-7: Preset Selection Screen



- Each preset has a 'Name' and a serial number 'Pn' (n=1, 2,...).
- A preset that was selected to be activated during the power-up procedure (see [Section 4.4.1.2.2](#)) is marked with '**(Power Up)**' near its serial number.
- Press shortly on one of the existing Preset to close this screen and activate it.
- In order to modify, delete or add a Preset, press and hold it until the 'Preset configuration' screen is displayed (see [Section 4.4.2](#)).

4.3.8 Selecting an Axis item (Target, Filter or Source)

- A button (Figure 4-5, item 7) is added for each Axis which is installed in the SR-800N system (located at the lower side of the main Absolute or Differential mode screen).



- Each button has 2 sections:
 - The upper section indicates the name of the current selected item in this Axis.
 - The lower shaded section indicates the name of the Axis.
 - The operator can name each Axis with meaningful value, for ease of use (see Section 4.4.3.4).
 - When there is an encoder position feedback, the actual position readout is added to the name of the item in brackets.



For example:

- Each Axis may be a Targets wheel, Filters Wheel or a Source Stage.

4.3.8.1 Axis Selection screen

- To invoke the 'Axis Selection screen' press one of the axis buttons at the main screen (Figure 4-5, item 7):

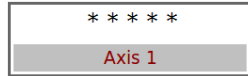
Figure 4-8: Axis Selection screen

 Home		 Back	 Setting	Axis1 Selection	
PH 1.05 mrad T1		PH 0.765 mrad T2		PH 0.380 mrad T3	
PH 3.333 mrad T4		PH 0.333 mrad T5		PH 0.400 mrad T6	
PH 0.433 mrad T7		PH 0.533 mrad T8		PH 0.567 mrad T9	
PH 0.600 mrad T10		PH 0.633 mrad T11		PH 0.667 mrad T12	
PH 0.700 mrad T13		PH 0.733 mrad T14		PH 0.767 mrad T15	

- The content of each item at the Axis Selection screen reflects its type:

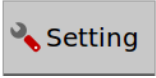
Type	Upper field	Lower field
Targets Wheel	The Name of the target	Serial number Tnn (T1 , T2)
Filters Wheel	The Name of the Filter	Serial number Fnn (F1 , F2)
Sources Stage	The Name of the Source	Serial number Snn (S1 , S2)

- The current active item is shaded.
- Press shortly on one of the available items to select it and close this screen. During the motion of the Axis, until it is stable in the required position, the Axis button at the main **Absolute** or **Differential** mode screen indicates



, with changing number of '*'.

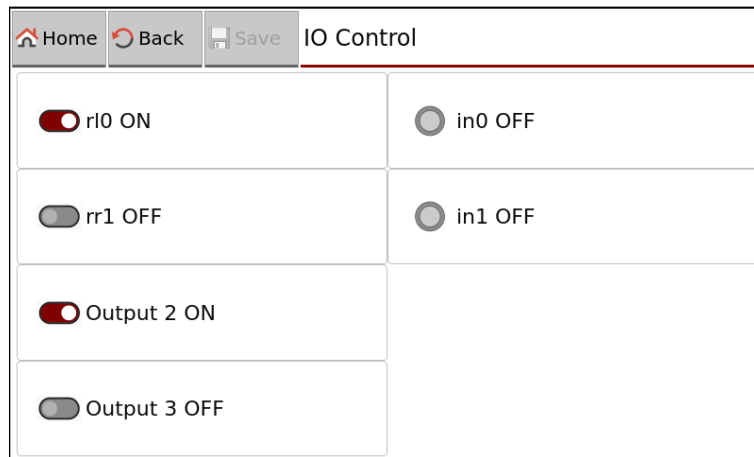
When it is stable, the name of the selected item will be displayed in the upper section of the Axis button.

- Press the  button at the top of the screen to modify the parameters of this axis, or to enable/disable its operation using the 'Axis Setting' menu Screen (see [Section 4.4.3](#)).
- Press and hold a button to modify the parameters of this item, or to delete it or to add a new one (of the same type) using the 'Configuration Screen' (see [Section 4.4.4](#)).

4.3.9 I/O Control Screen

- To invoke the I/O Control Screen press the  button (located at the upper side of the main Absolute or Differential mode screen, if this Option is installed):

Figure 4-9: I/O Control Screen



- Each output has an 'ON/OFF' switch (see [Section 4.3.2.2](#)) which enables the operator can change its state (ON or OFF):

Table 4-1: I/O Control Screen - Output activation meaning

Output type			Note
TTL	Logic '1'.	Logic '0'.	
Relay	Contact closed	Contact Open	
Open Collector	GND	Open	

- The Value of signal at each of the inputs is displayed using a state indicator:

Table 4-2: I/O Control Screen - Input indication meaning

Input type			Note
TTL	Logic '1'.	Logic '0'.	
Opto-Coupler	More than 3V	Open or less than 3V	

- Press and hold Output button or Input indication to invoke the alphanumeric Entry screen (see [Figure 4-4](#)) to set or modify its name.

4.3.10 'Info' screen

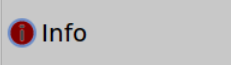
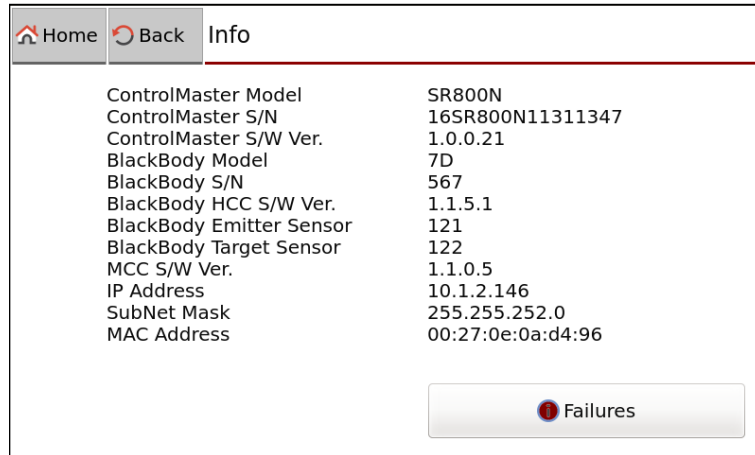
- To invoke the 'Info' screen press the  button (located at the upper right side of the main Absolute or Differential mode screen):

Figure 4-10: 'Info' screen



- The following information is displayed in the 'Info' screen are:

Table 4-3: 'Info' screen- Fields list

Field name	Description	Note
ControlMaster Model	Type of this ControlMaster	
ControlMaster S/N	The S/N of this ControlMaster	
ControlMaster S/W ver.	The S/W version of the embedded SBC	
Blackbody Model	Size (2, 4, 7..) and type (A for absolute, D for differential)	
Blackbody S/N	The S/N of the Blackbody Head	
Blackbody HCC S/W ver.	The S/W version of the HCC	
Blackbody Emitter sensor	S/N of the currently installed emitter temperature sensor	
Blackbody Target sensor	S/N of the currently installed target temperature sensor	
MCC S/W ver.	The S/W version of the MCC (if installed)	
IP Address	The IP address of this ControlMaster	
Subnet Mask	The subnet mask of the IP address of this ControlMaster	
MAC Address	The MAC address of this ControlMaster	

- If there are detected failures in installed components of the SR-800N system, the 'Failures' button is active with indicator , similar to the one in the 'Info' button.
- Refer to [Section 5.2](#) for information regarding the Failures information and troubleshooting.

4.3.11 LT Blackbody Head Operation

When LT Blackbody Head is used with the Julabo Circulation System, the following steps should be carried out:

4.3.11.1 Powering On the Julabo Circulation System

- Refer to the Julabo Circulation System Operation manual for further information.
- Verify that the liquid tank of the Julabo Circulation System is filled up to 30mm below the tank rim, fill it if needed (refer to '*Julabo Circulation System Operation manual*').
- Press the Circulator power switch and turn it ON (see [Figure 2-5](#)).
- Press the Controller power switch and turn it ON (see [Figure 2-5](#)).
- Turn On the SR-800N (see [Section 4.1](#))
- Verify correct fluid level after about 1 minute of operation; fill it if needed (refer to '*Julabo Circulation System Operation manual*').

4.3.11.2 Setting the Temperature

- Set the required temperature (see [Section 4.3.6](#))
- The SR-800N updates the liquid temperature set point of the Julabo Circulation System according to [Table 4-4](#):

Table 4-4: Julabo Circulation System Temperature Setting

SR-800N Set point range [°C]	Julabo Circulation System Set point [°C]	Note
-40 to -20	-30	
-20 to 0	-10	
0 to 40	20	
40 to 80	60	
80 to 150	100	

- The SR-800N waits until the liquid temperature of the Julabo Circulation System it is stable at its set point.
- The SR-800N stabilizes the temperature of the emitter to the required 'Set point' value.

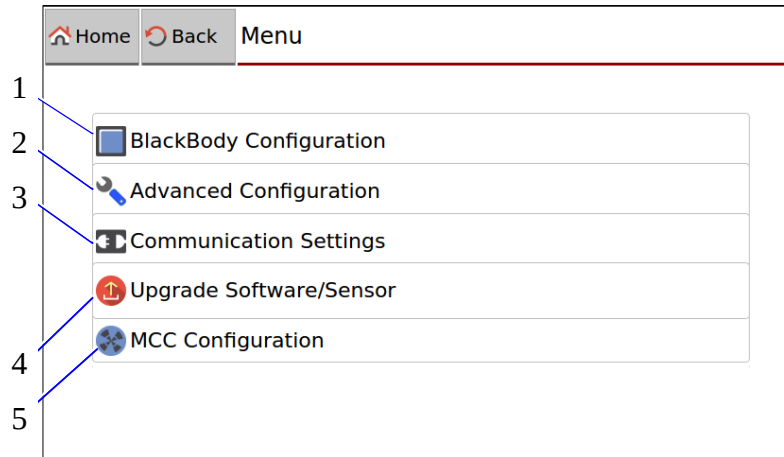
4.4 Configuration Procedures

The SR-800N configuration includes the following:

4.4.1 Main Menu screen

To invoke the Main Menu screen press the  button (located at the upper left side of the main **Absolute** or **Differential** mode screen):

Figure 4-11: Main Menu screen



The Main Menu screen has the following items:

- [Item 1](#) - a button for activating the 'Blackbody Configuration' screen (see [Section 4.4.1.1](#)).
- [Item 2](#) - a button for activating the 'Advanced Configuration' screen (see [Section 4.4.1.2](#)).
- [Item 3](#) - a button for activating the 'Communication Settings' screen (see [Section 4.4.1.3](#)).
- [Item 4](#) - a button for activating the 'Upgrade Software/Sensor' screen for handling software upgrade and new calibration information for the temperature sensors (see [Sections 5.5.5](#) and [5.6](#)).
- [Item 5](#) - Button for activating the 'MCC configuration' screen (see [Section 4.4.1.4](#)).

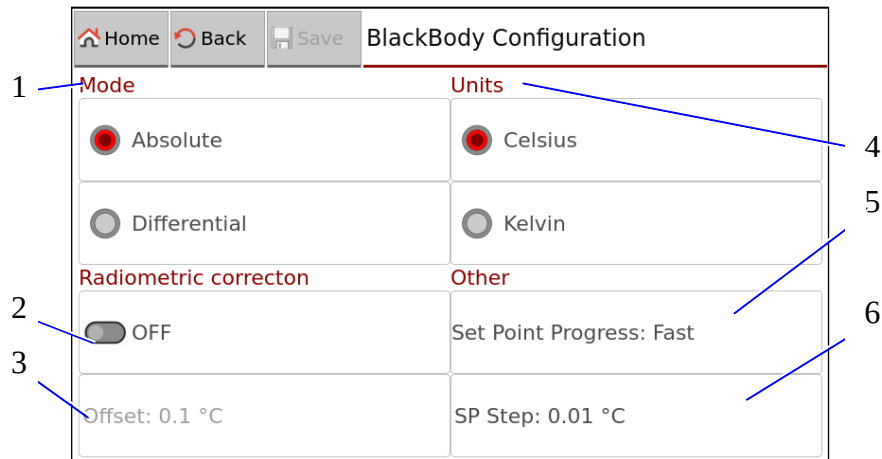
NOTE

This button is visible only if the MCC is installed.

4.4.1.1 Blackbody Configuration screen

To invoke the Blackbody Configuration screen press the  button in the Main Menu screen (Figure 4-11, Item 1):

Figure 4-12: Blackbody Configuration screen



The screenshot shows the 'BlackBody Configuration' screen. At the top are three buttons: 'Home' (house icon), 'Back' (circular arrow), and 'Save' (floppy disk icon). The screen is divided into two columns. The left column has a 'Mode' section with 'Absolute' (selected with a red dot) and 'Differential' (unselected with a grey dot). Below this is a 'Radiometric correcton' section with a toggle switch set to 'OFF'. At the bottom of the left column is an 'Offset: 0.1 °C' field. The right column has a 'Units' section with 'Celsius' (selected with a red dot) and 'Kelvin' (unselected with a grey dot). Below this is an 'Other' section with 'Set Point Progress: Fast' and 'SP Step: 0.01 °C'. Numbered callouts point to: 1. Mode section, 2. Radiometric correcton toggle, 3. Offset field, 4. Units section, 5. Kelvin option, and 6. SP Step field.

The possible functions in this Blackbody Configuration screen are:

- **Item 1** - Select the mode of operation – Absolute or Differential (if applicable) (see [Section 4.4.1.1.1](#)).
- **Item 2** - Activate/Inhibit the 'Radiometric Offset' Correction (if applicable).
- **Item 3** - Set the 'Radiometric Offset' value (if active and applicable).
- **Item 4** - Select the units of the measured temperature between Celsius and Kelvin (see [Section 4.4.1.1.2](#)).
- **Item 5** - Set the sensitivity of the Increment and Decrement temperature buttons.
This parameter affects the temperature set point change rate when pressing continuously the  or  buttons in the main screen (see [Section 4.4.1.1.3](#)).
- **Item 6** - Set the step size of the Increment and Decrement temperature buttons.
This parameter affects the temperature set point change rate when pressing the  or  buttons in the main screen (see [Section 4.4.1.1.4](#)).

NOTE

Options that are not relevant to the system
are disabled, and cannot be modified.

4.4.1.1.1 Mode of Operation

This control ([Figure 4-12, Item 1](#)) enables the Operator to set the Mode of Operation using 'Radio buttons' (see [Section 4.3.2.1](#)) - 'Absolute' or 'Differential'.

NOTE

The 'Differential' option is active only if the Blackbody Head is configured as 'Differential' device, and has a 'Target' temperature sensor.

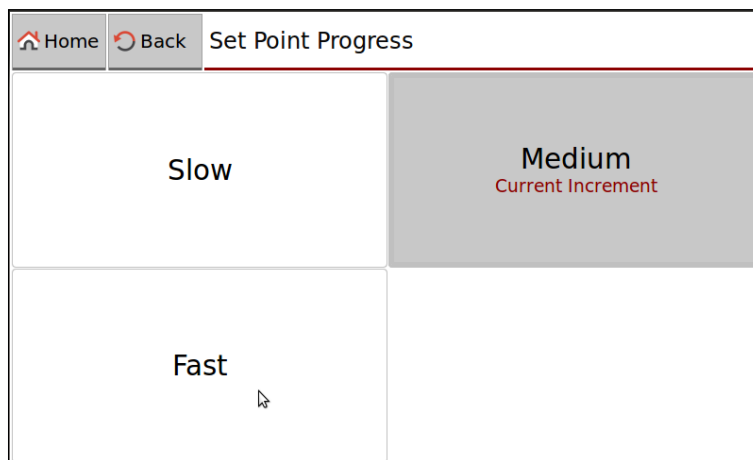
4.4.1.1.2 Temperature Measurement Units

This control ([Figure 4-12, Item 4](#)) enables the Operator to set the Temperature Measurement Units using 'Radio buttons' (see [Section 4.3.2.1](#)) - 'Celsius' or 'Kelvin'.

4.4.1.1.3 Set Point Progress setting screen

Pressing the 'Set Point Progress' value ([Figure 4-12, Item 5](#)) invokes the Set Point Progress setting screen:

Figure 4-13: Set Point Progress setting screen



- The current active sensitivity is designated.
- Press on one of the available values ('Slow', 'Medium' or 'Fast') to select it.

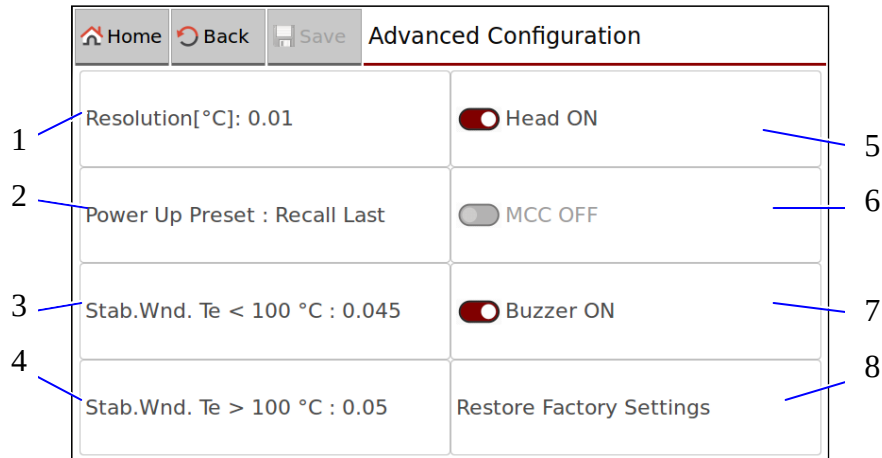
4.4.1.1.4 Set Point Step size setting screen

Pressing the 'SP Step' value ([Figure 4-12, Item 6](#)) invokes the numeric data entry screen (see [Figure 4-3](#)) to enter a new 'Set Point step size' value.

4.4.1.2 Advanced Configuration screen

To invoke the Advanced Configuration screen press the  button in the Main Menu screen ([Figure 4-11, Item 2](#)):

Figure 4-14: Advanced Configuration screen



The possible functions in this Advanced Configuration screen are:

- [Item 1](#) - Set the resolution of the temperature setting and readout (see [Section 4.4.1.2.1](#)).
- [Item 2](#) - Select the behavior of the SR-800N during power-up (see [Section 4.4.1.2.2](#)).
- [Item 3](#) - Set the size of the stabilization window for absolute set point below 100°C (see [Section 4.4.1.2.3](#)).
- [Item 4](#) - Set the size of the stabilization window for absolute set point above 100°C (see [Section 4.4.1.2.3](#)).
- [Item 5](#) - Turn On/Off the temperature control of the Blackbody Head (see [Section 4.4.1.2.4](#)).
- [Item 6](#) - Turn On/Off the Axes control; if applicable (see [Section 4.4.1.2.5](#)).
- [Item 7](#) - Turn On/Off the Buzzer of the ControlMaster (see [Section 4.4.1.2.6](#)).
- [Item 8](#) - Restore the factory settings (see [Section 4.4.1.2.7](#)).

NOTE

Options that are not relevant to the system
are disabled, and cannot be modified.

4.4.1.2.1 Temperature precision setting screen

- Press the 'Precision' value (Figure 4-14, Item 1) to invoke the Temperature precision setting screen:

Figure 4-15: Temperature precision setting screen

Home	Back	Temperature Precision	
1		0.1	
0.01		0.001 Current Precision	

- The current active precision is designated.
- Press on one of the available values ('1', '0.1', '0.01' or '0.001') to select it.

NOTE

The selected value is stored in the system configuration parameters and will be used at all proceeding operation.

4.4.1.2.2 Power-Up Preset setting screen

Press the 'Power Up Preset' value (Figure 4-14, Item 2) to invoke the Power-Up Preset setting screen:

Figure 4-16: Advanced Configuration screen - Power-Up Preset setting screen

Home	Back	Power Up Preset	
1.Abs_25		2.Diff_23	
No Preset Current Choice			

- The current active 'Preset' is designated.
- Press on one of the available 'Presets' or 'No Preset' to select it.

NOTE

The selected value is stored in the system configuration parameters and will be used at all proceeding operation.

When 'No Preset' is selected the SR-800N uses the **last** mode and temperature set-point.

4.4.1.2.3 Stabilization Window setting screen

- Press the 'Te<100°C' (Figure 4-14, Item 3) to invoke the Stabilization Window setting screen for temperature below 100°C, using the numeric data entry screen (see Figure 4-3).
- Press the 'Te>100°C' (Figure 4-14, Item 4) to invoke the Stabilization Window setting screen for temperature above 100°C, using the numeric data entry screen (see Figure 4-3).
- Pay attention to the range of valid values (the minimal and maximal values are displayed (Figure 4-3, Item 5)).
- The entered value is translated into '±' window around the 'Set point': for example a value of 0.015°C mean that the stabilization window is <Set point - 0.015> to <Set point + 0.015>

NOTE

The stabilization windows are defined for the Absolute temperature of the Emitter, regardless of the mode of operation ('Absolute' or 'Differential').

4.4.1.2.4 Turning On/Off the Blackbody Head Temperature control

Turning On/Off the Blackbody Head Temperature control is done using the control switch (Figure 4-14, Item 5), which is an 'ON/OFF Switch' (see Section 4.3.2.2).

NOTE

This function turns ON/OFF the temperature control of Blackbody head, while keeping the current mode of operation and temperature set-point.

4.4.1.2.5 Turning On/Off the Axes control

Turning On/Off the Axes control is done using the control switch (Figure 4-14, Item 6), which is an 'ON/OFF Switch' (see Section 4.3.2.2).

NOTE

This function is active only if MCC and Axes are installed in the SR-800N.

This function affects **all** the installed Axes.

4.4.1.2.6 Turning On/Off the Buzzer

Turning On/Off the Buzzer is done using the control switch (Figure 4-14, Item 7), which is an 'ON/OFF Switch' (see Section 4.3.2.2).

4.4.1.2.7 Restore the factory settings

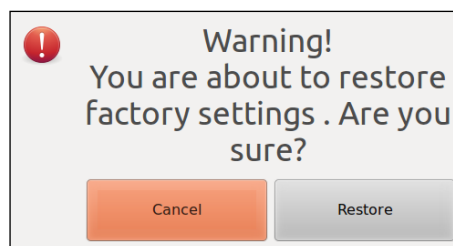
- In order to Restore the factory settings, the operator should press the button (Figure 4-14, Item 8).

NOTE

This function clears **all** the changes that were made in the 'Blackbody configuration' and 'Advanced Configuration' screens, and loads the value that were set during the Calibration and test procedures at CI Systems.

- A warning message pops-up. The operator should press the 'Restore' button to proceed with this function, or the 'Cancel' button to abort and keep the current values.

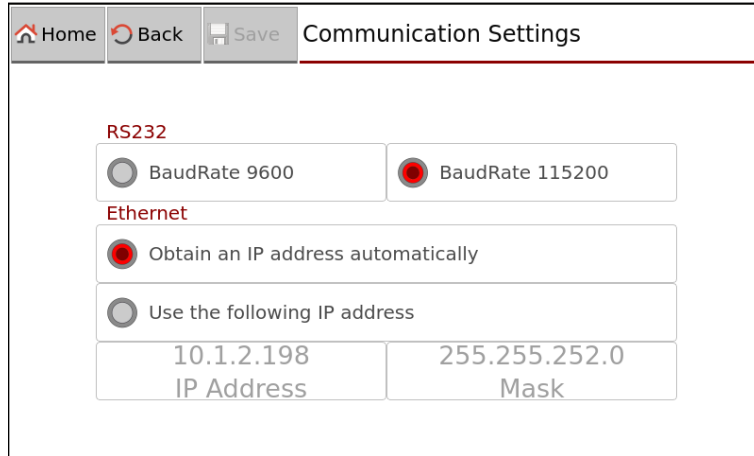
Figure 4-17: Restore the factory settings warning message



4.4.1.3 Communication Settings screen

- To invoke the Communication Settings screen press the  button in the Main Menu screen ([Figure 4-11, Item 3](#)):

Figure 4-18: Communication Settings screen



- Select baud rate for the serial communication (RS232).

NOTE

The default baud rate for SR-800N system is 115,200 bps.

The lower rate (9600 bps) should be only when SR-800/R remote application is used.

- Select between Dynamic ('automatic') and static IP address for the LAN communication of the ControlMaster.
- The selected mode is designated with .
- If static IP address is selected, press and hold the 'IP Address' and/or the 'Mask' field to updated their values.

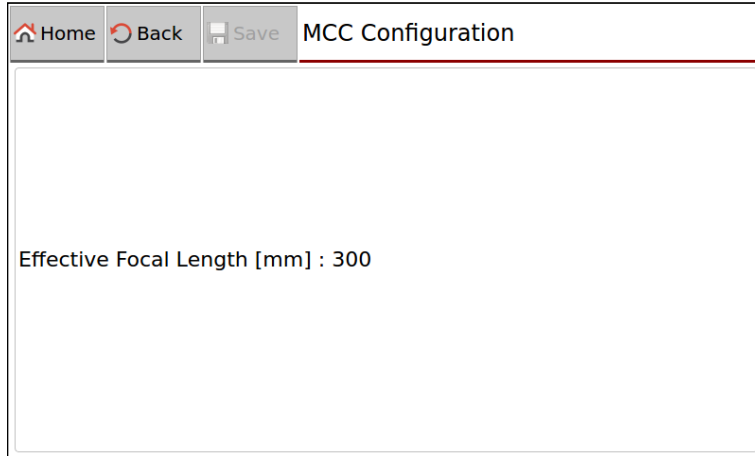
NOTE

The actual IP address of the ControlMaster is displayed in the 'Info' screen (see [Section 4.4.2.1](#)).

4.4.1.4 MCC Configuration screen

- To invoke the MCC Configuration screen press the  button in the Main Menu screen ([Figure 4-11, Item 5](#)):

Figure 4-19: MCC Configuration screen



- Press the 'Effective Focal Length' to enter a new value using the numeric data entry screen (see [Figure 4-3](#)).

NOTE

The 'Effective Focal Length' (EFL) value is used to create the 'Name' field of a standard target type.

4.4.2 Preset Configuration screen

- The Preset Configuration screen enables the operator to modify or delete existing Preset, or to add a new one.

Figure 4-20: Preset Configuration screen

- The selected Preset serial number is displayed at the bottom
- The screen includes all the parameters that configure this Preset
Each parameter is displayed as a button with 2 sections:
 - The upper section displays the current value of this parameter.
 - The lower section displays the name of this parameter.
- The following table describes the available parameters:

Table 4-5: Preset Configuration screen- parameters list

Item	parameter	Description	Reference	Note
1	Preset Name	Free text field, describing the 'Preset'	4.4.2.1	
2	Blackbody Set Point	A temperature Set Point	4.4.2.2	Must be filled
3	Blackbody mode	Absolute or Differential (if applicable)	4.4.2.3	Must be filled
4	Axis 1	Selected item from Axis 1	4.4.2.4	Visible if Axis 1 is installed

- If other Axes (2, 3 or 4) are installed, their activation button will be visible, similar to one of Axis 1.
- Press each field to modify its content.
- Use the navigation buttons at the bottom to delete this 'Preset', to add a new one or to move to the next/previous (see [Section 4.3.1](#)).

4.4.2.1 Preset Name Editing

- Press and hold the 'Preset Name' field to modify or set a new 'Name' value using the alphanumeric entry (Figure 4-4).

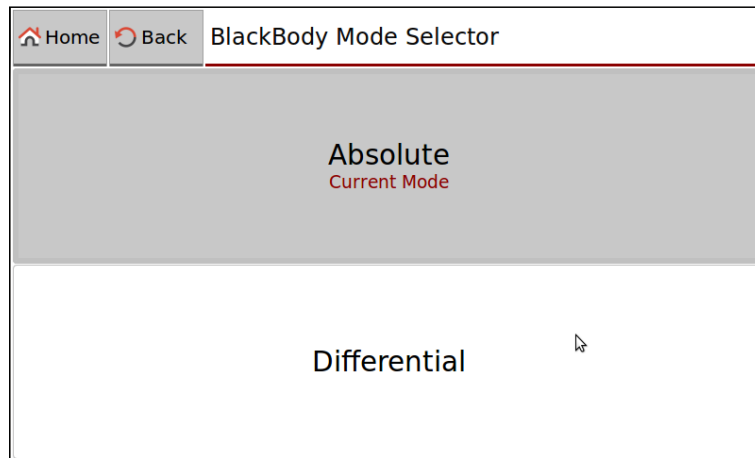
4.4.2.2 Preset 'Set Point' value

- Press and hold the 'Set Point' field to modify or set its value using the numeric entry screen (see Figure 4-3) is displayed.

4.4.2.3 Preset 'Mode' configuration Screen

- Press and hold the 'Mode' field to invoke the Preset 'Mode' configuration Screen.

Figure 4-21: Preset 'Mode' configuration Screen



- The current active sensitivity is designated.
- Press on one of the available values ('Absolute' or 'Differential') to select it.

NOTE

The 'Differential' option is active only if the Blackbody Head is configured as 'Differential' device, and has a 'Target' temperature sensor.

4.4.2.4 Preset 'Axis 1' configuration

- Press and hold the 'Axis 1' field to select an item in 'Axis 1' using the 'Axis 1 Selection' screen (see Figure 4-8), which will be operated in this Preset. For example – if 'Axis 1' is a Targets wheel, this option enables to select a target, which will be selected and activated in the current 'Preset'.
- This is an optional field, and will be visible only if 'Axis 1' is installed and available.

4.4.3 Axis Setting screen

- The Axis Setting screen enables the operator to modify existing Axis (Target, Filter or Source).

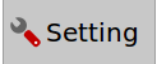
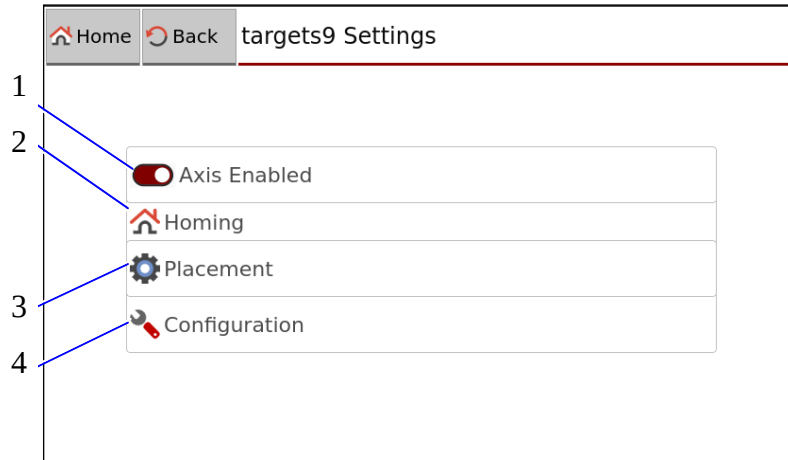
- To invoke this screen press the  button (located at the Upper section of the 'Axis Selection' screen, see [Figure 4-8](#)):

Figure 4-22: Axis Setting screen



- The operator can use this screen for:
 - [Item 1](#) - Enable/disable the position control of this Axis (see [Section 4.4.3.1](#))
 - [Item 2](#) - Force 'Homing' movement to verify the current position or clear accumulative positioning errors (see [Section 04.4.3.2](#)).
 - [Item 3](#) - Verify or modify the 'Target Plates' placement in the 'Targets Wheel' (see [Section 4.4.3.3](#))
 - [Item 4](#) - Modify the configuration parameters of this Axis (see [Section 4.4.3.4](#))

4.4.3.1 Enable/Disable the Axis control

Enable/Disable the Axis control is done using the control switch ([Figure 4-22, Item 1](#)), which is an 'ON/OFF Switch' (see [Section 4.3.2.2](#)).

NOTE

This function affects only this Axis.

4.4.3.2 Homing

Forced Homing is done using  button (Figure 4-22, Item 2). The Axis will rotate to its 'Home' (reference) position and back to the last selected position.

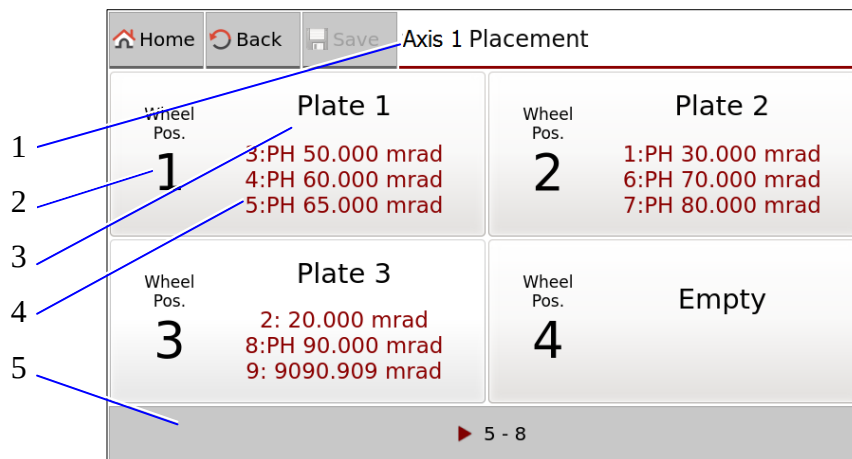
NOTE

When encoder position feedback is used this function is not active.

4.4.3.3 Axis Placement screen

- Press the  in the 'Axis Setting' screen (see Figure 4-22, Item 3) to invoke the Axis Placement screen.
- In case of Targets Wheel, this screen enables the operator to define which Targets Plate is located at each 'Position' of the Targets wheel (see Appendix A – Targets Wheel for more information).

Figure 4-23: Axis Placement screen



- This screen displays the current placement of Targets Plate in the selected Axis
- This screen contains up to 4 'Target wheel Positions' with the following information:
 - **Item 1** - indicates the name of the selected Axis.
 - **Item 2** - the serial number of each 'Position' in the Targets Wheel.
 - **Item 3** - the serial number of the Target Plate which is currently placed in this 'Position'.
 - **Item 4** - a list of all the Targets which are included in this Targets Plate.

- **Item 5** - indicates the total number of available Positions (according to the factory basic initialization), and enables the operator to navigate between the screens which displays the other 'Positions'.

NOTE

If the list of Target included I a specified 'Target Plate' is not correct, check the parameters of the specified Targets, and correct value in the 'Plate' field (see [Section 4.4.4](#)).

- In order to modify (replace or remove) the Targets Plate that is installed in a position, press and hold the 'Position' until the 'Plate Selection' screen will be displayed (see [Section 4.4.3.3.1](#)).

4.4.3.3.1 Targets Plate Selection screen

- Press and hold a 'Target wheel Positions' (see [Figure 4-23](#)) to invoke the Targets Plate Selection screen.

Figure 4-24: Targets Plate Selection screen

Home Back		Targets : Choose Plate for Wheel Pos. 1
Plate 1 Current Choice	Plate 2	Plate 3
Plate 4	Plate 5	Plate 6
Plate 7	Plate 8	(Empty)

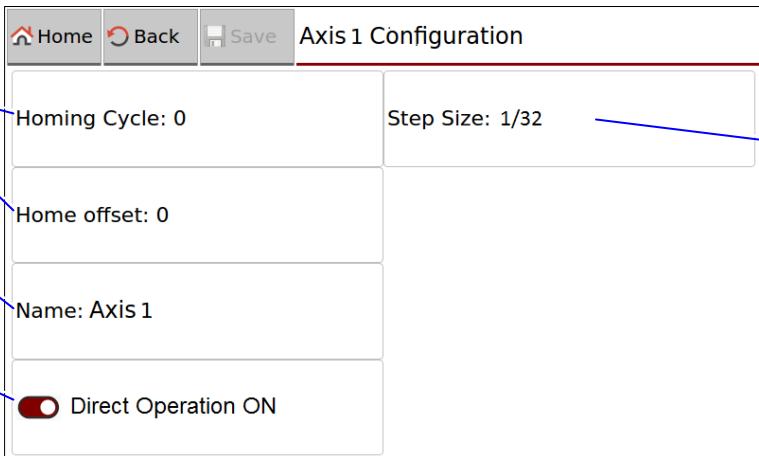
- The header of this screen indicates the Serial Number of the selected 'Position' in the selected Axis (the name of the Axis is displayed).
- The current selected 'Target Plate' is marked ('**Current Choice**').
- Press on one of the available 'Targets Plate' to modify the selection.
- If the current 'Position' is empty or not used, select the '**(Empty)**' option.
- After selecting a Plate Number, this screen is closed and the 'Axis Placement screen' is displayed.

- The operator should verify the list of targets that are included in the selected 'Targets Plate' (see [items 2-4](#) in [Figure 4-23](#)).

4.4.3.4 Axis Configuration screen

- Press the  in the 'Axis Setting' screen (see [Figure 4-22, Item 4](#)) to invoke the Axis Configuration screen.

Figure 4-25: Axis Configuration screen



The screenshot shows the 'Axis 1 Configuration' screen. At the top, there are three buttons: 'Home' (with a house icon), 'Back' (with a circular arrow icon), and 'Save' (with a floppy disk icon). The screen is divided into several sections. On the left, there are four numbered callouts: 1 points to the 'Homing Cycle: 0' field, 2 points to the 'Home offset: 0' field, 3 points to the 'Name: Axis 1' field, and 4 points to a toggle switch labeled 'Direct Operation ON'. On the right, there is a field for 'Step Size: 1/32' with a callout 5 pointing to it.

- The operator can use this screen for:
 - [Item 1](#) - Set or modify the 'Homing Cycle' value of this Axis (see [Section 4.4.3.4.1](#)).
 - [Item 2](#) - Set or modify the 'Home Offset' value of this Axis (see [Section 4.4.3.4.2](#)).
 - [Item 3](#) - Set or modify the name of this Axis (see [Section 4.4.3.4.3](#)).
 - [Item 4](#) - Enable/disable the position control of this Axis (see [Section 4.4.3.4.4](#)).
 - [Item 5](#) - Set or modify the step size of the motor of this Axis (see [Section 4.4.3.4.5](#)).

4.4.3.4.1 Setting the 'Homing Cycle' value

- Tap on the 'Homing Cycle' field ([Figure 4-25, Item 1](#)) to modify its value using the 'Numeric Entry Screen' (see [Figure 4-3](#)).
- The 'Homing Cycle' value meaning is that the SR-800N will perform 'Homing Cycle' after the specified number of movements.
For example, if the 'Homing Cycle' value is 10, at each 10th movement of the Axis, it will move first to its 'Home' position before moving to the required position.

NOTE

The value '0' means that there will be no homing between positioning activities.

(See [Appendix A – Targets Wheel](#) for more information)

4.4.3.4.2 Setting the 'Home Offset' value

- Tap on the 'Home Offset' field ([Figure 4-25, Item 2](#)) to modify its value using the 'Numeric Entry Screen' (see [Figure 4-3](#)).
- 'Home Offset' value is used for calibrating the 'Home' position of the Axis.

NOTE

Changing this value **affects** the positioning of all the items in this Axis.

(See [Appendix A – Targets Wheel](#) for more information)

4.4.3.4.3 Setting the 'Name' of the Axis

- Tap on the 'Name' field ([Figure 4-25, Item 3](#)) of the Axis to modify it using the 'Alphanumeric Entry Screen' (see [Figure 4-4](#)).

NOTE

The name of the Axis will be displayed on the activation button in the main screen (see [Figure 4-5, item 7](#)) and at the heading of all the screens that operate and configured it (see [Figure 4-8, Figure 4-22](#) etc.)

4.4.3.4.4 Enable/Disable 'Direct Operation'

- Enable/Disable 'Direct Operation' is done using the control switch (Figure 4-25, Item 4), which is an 'ON/OFF Switch' (see Section 4.3.2.2).
- When this function is active, a new button item is added to the 'Axis Selection' screen, name 'Direct Operation' (see Section 4.4.5).

4.4.3.4.5 Step Size Selection screen

- Tap on the 'Step Size' field (Figure 4-25, Item 5) to invoke the Step Size Selection screen.

Figure 4-26: Step Size Selection screen

Home Back Step Size	
1/32	1/16
1/8 Current Choice	1/4
1/2 NC	1/2
1 NC	1

- The current selected 'Step Size' is marked ('**Current Choice**').
- Press on one of the available 'Step Size' to modify the selection and close this screen.

NOTE

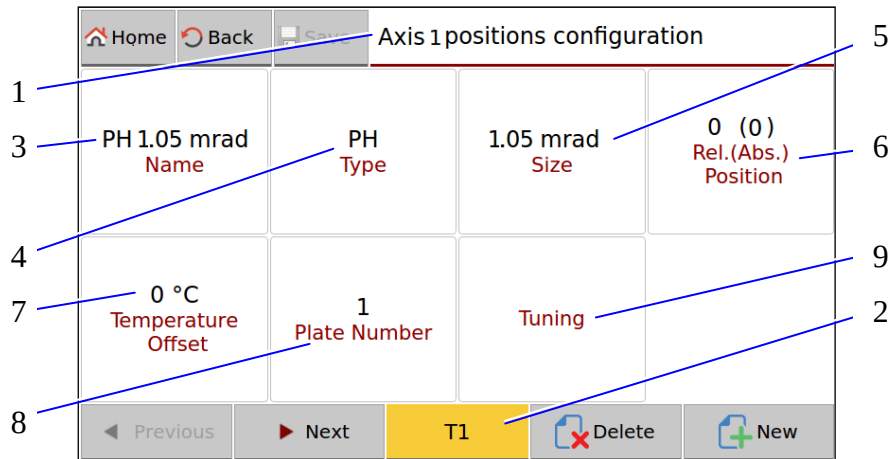
Changing this value affects the positioning of
all the items in this Axis.

Larger step size adds power, but with reduced resolution.

4.4.4 Axis item Configuration screen

- The Axis item Configuration screen enables the operator to modify or delete existing item (Target, Filter or Source) or to add a new one.
- To invoke this screen press and hold an item in the 'Axis selection' screen (see Figure 4-8).

Figure 4-27: Axis item Configuration screen



- **Item 1** - The Header of this screen indicates the name of the selected Axis
- **Item 2** – Displays the serial number of the selected item (T1, T2... for targets, F1, F2....for Filters and S1, S2....for Sources).
- Each parameter of the selected item is displayed as a button with 2 sections:
 - The upper section displays the current value of this parameter.
 - The lower section displays the name of this parameter.
- The following table describes the available parameters:

Table 4-6:Axis item Configuration screen- parameters list

Item	Name	Description	Reference	Note
3	Name	For standard type of item, build from Type, Size and EFL		Automatic
		For custom type of item	4.4.4.1	Must be filled
4	Type	Type of item – for example type of target	4.4.4.2	Must be filled
5	Size	When applicable - for example the size a specified type of target in mm.	4.4.4.3	For Targets only

Item	Name	Description	Reference	Note
6	Rel. Position	A numeric field, the position of this item in the motorized Axis, for Targets and Filters - <u>with respect to the center of its plate.</u> For Sources, - <u>with respect to the home /reference</u> position.	4.4.4.4	Must be filled
7	Temp. Offset	A numeric field, used for Test management. Indicates the temperature difference between the target wheel and the target.	4.4.4.5	
8	Plate number	The serial number of the plate, in which this item is installed. <u>Valid only for Targets and Filters</u>	4.4.4.6	Must be filled
9	Tuning	Invokes the Target Tuning screen which enables the operator to test and modify the Relative Position field value.	4.4.4.7	

- Press and hold any field to modify its content:
 - Free text fields will be modified using the alphanumeric entry screen (see [Figure 4-4](#)).
 - Numeric fields will be modified using the numeric entry screen (see [Figure 4-3](#)).
- Use the navigation buttons at the bottom to delete this item, to add a new one or to move to the next/previous (see [Section 4.3.1](#)).

4.4.4.1 Item 'Name' setting

- For items that their 'Type' value is not 'Custom', this field is built automatically from their 'Type', 'Size' divided by Effective Focal Length (EFL) value (see [Section 04.4.1.4](#)).
- For items that their 'Type' is 'Custom', press the 'Name' button ([Figure 4-27, Item 1](#)) to set or modify its name using the 'Alphanumeric Entry Screen' (see [Figure 4-4](#)).

4.4.4.2 Item 'Type' Selection Screen

- Press the 'Type' button ([Figure 4-27, Item 2](#)) to invoke the 'Type Selection' Screen.

Figure 4-28: Item 'Type' Selection Screen – Type of targets

Home Back Target Type Selector		
PH Current Choice	CROSS	BLANK
USAF	RECT	OPEN
4B	STEP	CUSTOM
SQR	SLIT	BRST

- The current selected 'Type' is marked ('**Current Choice**').
- Press on one of the available 'Types' to modify the selection and close this screen.

NOTE

The 'Custom' type should be used for type of items which are not included in this list.

4.4.4.3 Target Size setting

- Press the 'Size' button ([Figure 4-27, Item 3](#)) to modify its value (in mm) using the 'Numeric Entry Screen' (see [Figure 4-3](#)).

4.4.4.4 Item Position Setting

4.4.4.4.1 Relative Position Setting

- If the Axis is defined as 'Relative' position (factory setting), the 'Position' button indicated the Relative and Absolute (in brackets) position of the item in motor steps.
- This type of position is used in large 'Targets Wheel' where the Relative position is with respect to the center of the 'Targets Plate' that contains the target, while the 'Absolute' position is with respect to the 'Home' (reference) position of the Targets Wheel (see [Appendix A – Targets Wheel](#)).

- Press the 'Position' button ([Figure 4-27, Item 4](#)) to modify its Relative Position value (in steps) using the 'Numeric Entry Screen' (see [Figure 4-3](#)). The entered number can be positive or negative, within the specified limits.

4.4.4.4.2 Absolute Position Setting

- If the Axis is defined as 'Absolute' position (factory setting), the 'Position' button indicated only its Absolute position in motor steps.
- Press the 'Position' button ([Figure 4-27, Item 4](#)) to modify its Absolute Position value (in steps) using the 'Numeric Entry Screen' (see [Figure 4-3](#)). The entered number can is always positive, within the specified limits.

NOTE

If encoder feedback is used, the entered value
(Relative or Absolute) is in encoder counts.

4.4.4.5 Target Temperature Offset Setting

- Press the 'Temp. Offset' button ([Figure 4-27, Item 5](#)) to modify the Temperature Offset value using the 'Numeric Entry Screen' (see [Figure 4-3](#)).

4.4.4.6 Target Plate Number Setting

- Press the 'Plate' button ([Figure 4-27, Item 6](#)) to set or modify the Plate number that contain the current target using the 'Numeric Entry Screen' (see [Figure 4-3](#)).

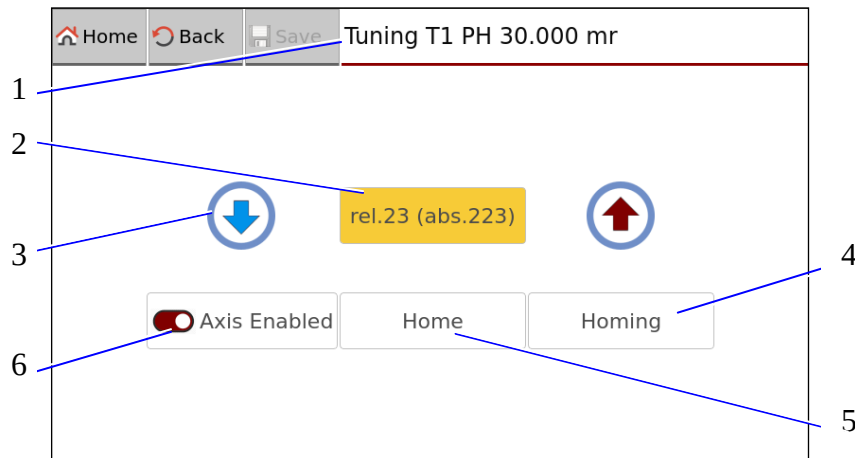
NOTE

This field is **essential** when working with Targets Wheel
with Relative position of the Targets.

4.4.4.7 Target Tuning screen

- Press the 'Tuning' button in the 'Target Configuration' screen of the selected target (see [Figure 4-27](#)) to invoke the Target Tuning screen.
- This screen enables the operator to test and fine tune the 'Relative position' of the Target (see [Appendix A – Targets Wheel](#) for more information).

Figure 4-29: Target Tuning screen

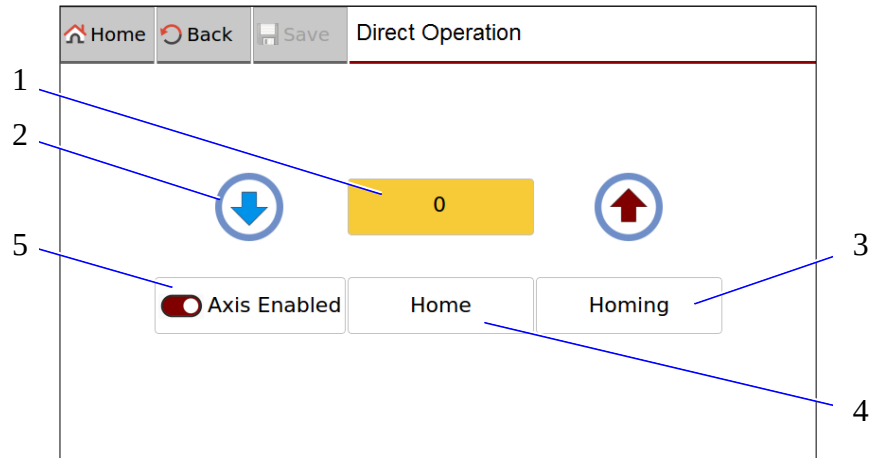


- **Item 1** - indicates the name of the selected Target.
- The operator can use this screen for:
 - **Item 2** - indicates the current Relative and Absolute (in brackets) position of the target. Tap on it to enter a new Relative value manually using the numeric entry screen (see [Figure 4-3](#)).
 - **Item 3** - (the  or  buttons) can be used to rotate the 'Target Wheel'. Each press moves it 1 step in the required direction and updates the 'Relative position' value (**Item 2**).
 - **Item 4** - Press the 'Homing' button to force 'Homing' movement – the 'Target Wheel' rotates to its 'Home' (reference) position and then back to the Target's position (used for verifying the new placement value).
 - **Item 5** - Press the 'Home' button to rotate the 'Target Wheel' to its 'Home' (reference) position.
 - **Item 6** – press the 'Axis Enabled' button which is an 'ON/OFF Switch' (see [Section 4.3.2.2](#)).
This function may be used for temporary disable the 'Targets Wheel' for maintenance (manual rotation for example). At the end of this activity, use the button to enable 'Targets Wheel'.
- Use the 'save' button to save the 'Relative position' value to Target's parameters and quit this screen.

4.4.5 Direct Operation screen

- The Direct Operation screen enables the operator to operate the axis directly without the need to define an item. For example, this function can be used for calibrating the 'Home' (reference) position of the Axis.
- To invoke this screen press the 'Direct Operation' item in the 'Axis selection' screen (see [Figure 4-8](#)).

Figure 4-30: Direct Operation screen



- The operator can use this screen for:
 - **Item 1** - indicates the current Absolute position of the Axis. Tap on it to enter a new value manually using the numeric entry screen (see [Figure 4-3](#)).
 - **Item 2** - (the  or  buttons) can be used to move the Axis. Each press moves it 1 step in the required direction and updates the 'position' value (**Item 1**).
 - **Item 3** - Press the 'Homing' button to force 'Homing' movement – the Axis moves to its 'Home' (reference) position and then back to the last position (used for verifying the new placement value).
 - **Item 4** - Press the 'Home' button to move the Axis to its 'Home' (reference) position.
 - **Item 5** – press the 'Axis Enabled' button which is an 'ON/OFF Switch' (see [Section 4.3.2.2](#)). This function may be used for temporary disable the Axis for maintenance (manual movements for example). At the end of this activity, use the button to enable the Axis.

5. Maintenance

This chapter provides safety and troubleshooting maintenance information, and step-by-step instructions on how to perform preventive and scheduled maintenance tasks on the SR-800N.

5.1 Preventive/Periodic Maintenance Tasks

The SR-800N Preventive/Scheduled maintenance includes the following tasks:

5.1.1 Visual Inspection

- Verify that there is no damage to the components and, cables and harnesses.
- Search for any loose part and tighten them (screws, connectors, etc.).
- Verify proper connection of all connectors.

5.1.2 Cleaning

- Wipe all SR-800N sub assemblies using a soft hairbrush.
- Clean all SR-800N subassemblies using a soft cloth, soaked in authorized cleaning material (IPA or similar).

5.1.3 Calibration

The calibration data of the temperature sensors of SR-800N Blackbody is stored in it.

The sensors shall be periodically calibrated (suggested - once a year).

The calibration is done by replacing the sensor with a new one, and updating the calibration data (see [Section 5.4](#))

5.2 Troubleshooting

This section includes description of the troubleshooting principle followed by a list of all-possible failure symptoms and instructions for locating them.

5.2.1 General Failures

Table 5-1: General Failures

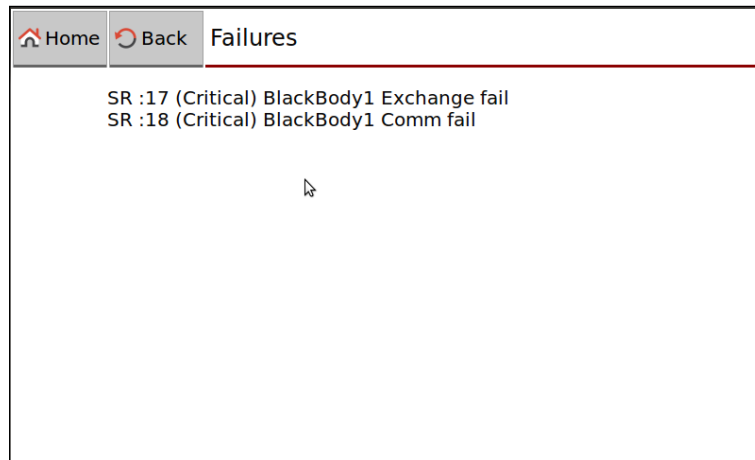
Failure	Possible Cause	Activity
SR-800N ControlMaster is not powering ON (Mains power switch is not illuminated)	1. No mains power 2. Power cord is not connected properly 3. The fuse was blown out 4. Internal power supply failure or Main power switch fault	1. Check main power 2. Verify power in cord is fully inserted and secured. 3. Check the fuse, replace if needed. 4. Replace the ControlMaster
SR-800N ControlMaster is not powering ON (Mains power switch is illuminated)	1. Internal power supply failure 2. PA failure 3. LCD or SBC failure	Replace the ControlMaster

5.2.2 Failures screen

- To invoke the Failures screen press the button at the lower right side of the 'Info' screen (see [Figure 4-10](#)):



Figure 5-1: Failures screen



NOTE

This screen pops up automatically if errors or failures are detected during the power-on procedure.

- Each line has the following structure
'Component' : 'Failure#('Severity')Failure description'
- 'Component' – the suspected item 'BB' for Blackbody head, 'SR' for ControlMaster or system.
 - 'Failure#' – the numeric code of the failure
 - 'Severity' – the severity level of the detected failure.
Those marked as 'Critical' disable the operation of the SR-800N.
 - 'Failure description' – general description of the detected failure.

5.2.2.1 Blackbody Failures messages

The following table describes the Failure messages codes related to the Blackbody head (Failure message that start with 'BB').

Table 5-2: Blackbody Failures messages

Code	Description	Possible Cause	Activity
0	Head Comm fail	Blackbody head internal communication or control card (TSC/HCC)	1. Check 'Head' harness is connected correctly on both ends. 2. Replace 'Head' harness 3. Restart SR-800N. if repeat send Blackbody head repair
1	BIT fail -	Blackbody head internal component failure	Restart SR-800N. if repeat send Blackbody head repair
8 9	Head data fail Head Tables fail	The Blackbody data tables are corrupted.	1. Restart SR-800N. 2. Refresh the head calibration data 3. if repeat send Blackbody head repair
12	Calibration Error	The auto calibration procedure failed	Restart SR-800N. if repeat send Blackbody head repair
13 14	Emitter Table error Emitter Table error	The Blackbody data tables are corrupted.	1. Restart SR-800N. 2. Refresh the head calibration data 3. if repeat send Blackbody head repair
16 17	Emitter Temp fail Target Tem error	The Blackbody head fail to read the temperature	1. Restart SR-800N. 2. Open the Temperature sensor cover, verify that the sensor wire is not damaged, and its connector is connected properly 3. If repeat send Blackbody head repair

5.2.2.2 ControlMaster Failure messages

The following table describes the Failure messages codes related to the ControlMaster or system (failure message that start with 'SR').

Table 5-3: Failures screen

Error Code	Error Message	Possible Cause	Activity
0	SR800 fail	This is a general failure bit.	Restart SR-800N.
1 17 18	Head 1 not exists Head 1 exchange fail Head 1 comm fail	The ControlMaster SBC failed to commentate with the HCC	Restart SR-800N. if repeat send ControlMaster head repair
7 17	PA Comm fail	At least one of the PAs in the Black ControlMaster failed	Restart SR-800N. if repeat send ControlMaster head repair
18	PA Init Fail	At least one of the PAs in the ControlMaster failed	Restart SR-800N. if repeat send ControlMaster head repair
21	Uniformity table fail	ControlMaster HCC failure	Restart SR-800N. if repeat send ControlMaster head repair
10	EEPROM fail	Blackbody head control card failure (HCC)	Restart SR-800N. if repeat send Blackbody head repair

5.3 Fuse Replacement



WARNING

The fuse is powered even when the SR-800N is OFF.

Before replacing the fuse, power-off the SR-800N and disconnect the power cord from the mains.

Failure to comply may result in injury to personnel and/or damage to equipment.

5.3.1 Procedure

- Power-Off the SR-800N.
- Disconnect the power cord from mains.
- Remove the fuse cover using a Flathead screwdriver.
- Remove the faulty fuse from the cover.
- place a new fuse in the cover:
 - Fuse rating: 10A, 250V
 - Fuse size: 20/5mm.
- Re-install the fuse cover and tighten using a Flathead screwdriver.
- Reconnect the power cord to mains.
- Power-On the SR-800N.

5.4 Temperature sensors Calibration methods

There are two methods to calibrate the temperature sensors of the SR-800N:

- Using a standalone calibration instrument.
- Using a removable temperature sensor.

5.4.1 Using a standalone calibration instrument

In this method, additional reference temperature sensor, connected to a standalone calibration instrument, is installed in the in the Emitter of the Blackbody Head.

The readout of both temperature sensors (the reference and the system) is compared through the entire operational temperature range.

The result is a Calibration Data table that is loaded into the SR-800N.

The Calibration Data table is stored in a non-volatile memory inside the Blackbody Head.

The Calibration data which is generated by this method is influenced by slight thermal gradients that exist across the Emitter between the positions of the sensors. This gradient is a function of the temperature of the Emitter (increases at high temperatures).

Using this method for calibrating a Target sensor is a bit complex because of the face that the temperature of the target plate cannot be controlled. For this, the reference sensor is installed in a copper block, stabilized at ambient temperature. Note that this method is only accurate for narrow temperature range near the ambient temperature.

The recommended stand-alone calibration instrument is CK-800.

Contact CI-Systems for further information regarding this device.

5.4.2 Using a removable temperature sensor

In this method, the existing temperature sensor is removed and replaced with a new calibrated one.

This method is based on the high accuracy, stability and repeatability of the temperature measurement circuit of the SR-800N, which allow changing calibrated temperature sensors in the Blackbody Head while maintaining calibration and traceability.

Each temperature sensor undergoes a standalone calibration procedure using a thermal bath at CI-Systems premises, with high traceability

The outcome of this procedure is a unique Calibration Data table, marked with a Serial Number and Calibration date, is etched into a USB mass storage device which is supplied with each calibrated temperature sensors.

The design of the Blackbody Head enables easy removal and installation

to support this method.

After installation of a new temperature sensor, its Calibration Data table is loaded into the SR-800N by connecting the USB mass storage device to the ControlMaster.

The Calibration Data table is stored in a non-volatile memory inside the Blackbody Head.

This method is accurate for both Emitter and Target temperature sensors

5.5 Temperature Sensors Maintenance

CAUTION

Shut down the power supply prior to any action taken.
Allow adequate time for the Emitter to cool down.

Avoid bending the sensor cable and the connections
between the cable and the sensor/USB connector.

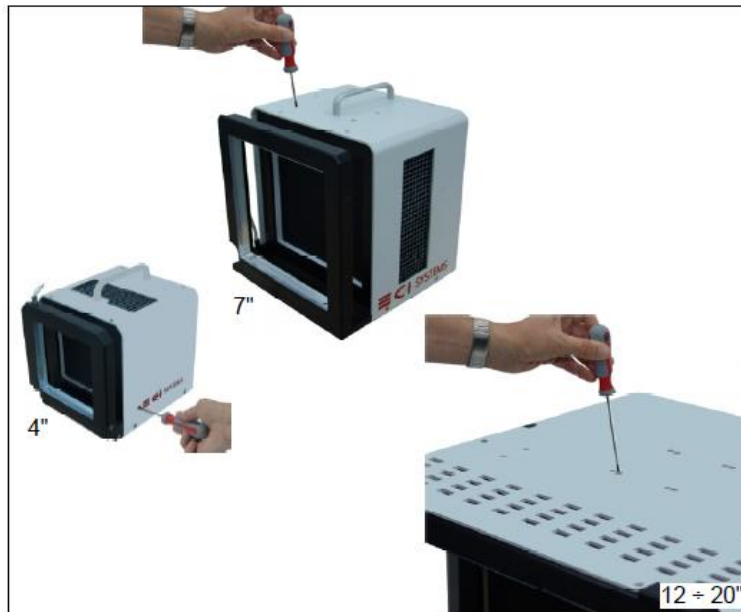
The sensor is fragile - take care.

5.5.1 4 ÷ 20" Blackbody Heads Emitter Sensor Replacement

5.5.1.1 Removal

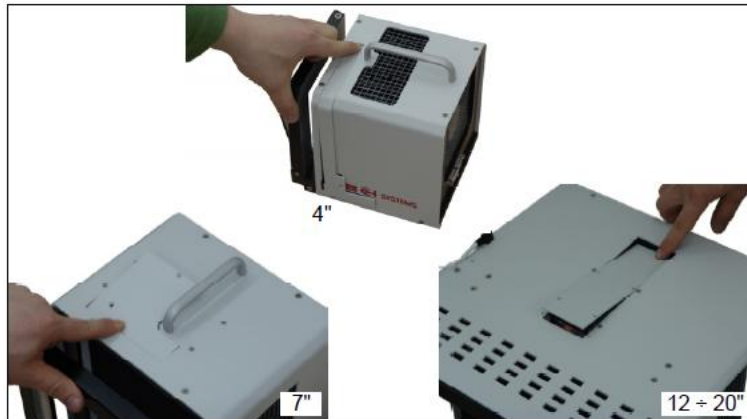
- Remove the screws securing the Temperature Sensor Cover:

Figure 5-2: 4 ÷ 20" Blackbody Heads Emitter Sensor Cover (1/2)



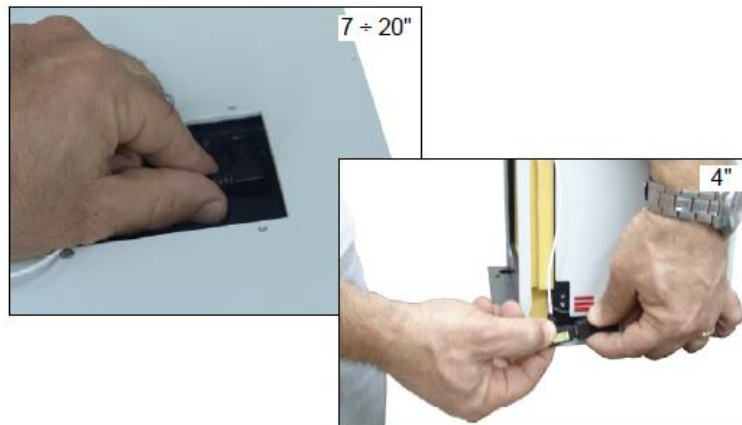
- Carefully remove the Sensor Cover by pressing on it:

Figure 5-3: 4 ÷ 20" Blackbody Heads Emitter Sensor Cover (2/2)



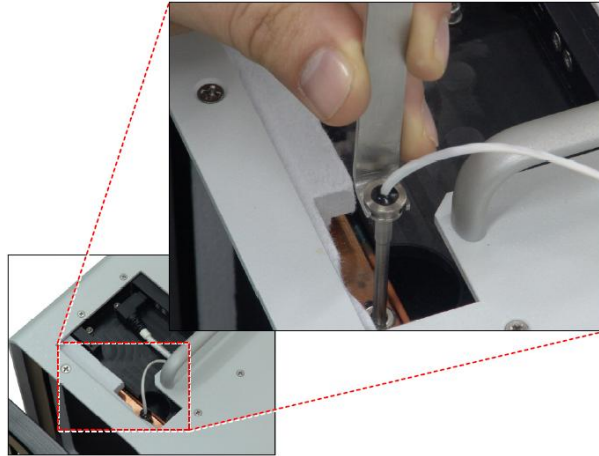
- Disconnect the Emitter Temperature Sensor connector from the Blackbody:

Figure 5-4: 4 ÷ 20" Blackbody Heads Emitter Sensor connector



- Pull out the Emitter Temperature Sensor from its port using the dedicated tool:

Figure 5-5: *4 ÷ 20" Blackbody Heads Emitter Sensor pulling out.*



NOTE

The Temperature sensor is held in place by built-in magnet; therefore, a reasonable force is required to separate the sensor from its port.

- Remove the Temperature sensor and store it in its package to avoid damage.
- Clean the Sensor port using the supplied plastic straw - Insert it into the Sensor port, rotate it several times and remove it.

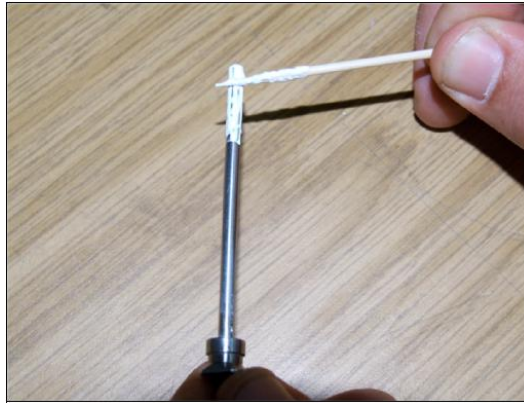
Figure 5-6: *4 ÷ 20" Blackbody Heads Emitter Sensor Port Cleaning*



5.5.1.2 Installation

- Apply a thin layer of thermal grease (supplied with the kit) on the bottom section of the Temperature sensor (approx. 1cm on the free end of the sensor), using a toothpick:

Figure 5-7: Greasing the Sensor



- Insert the Temperature sensor into the Emitter port until the magnet secures it.
- Connect the Temperature Sensor connector to the Blackbody (see [Figure 5-4](#)).
- Arrange the Temperature sensor's wiring so that the installing the Sensor cover will not damage it.
- Place the Sensor Cover and tighten its fasteners (see [Figure 5-2](#)).

NOTE

When placing the sensor cover ensure that there is no obstruction and the cover closes easily.

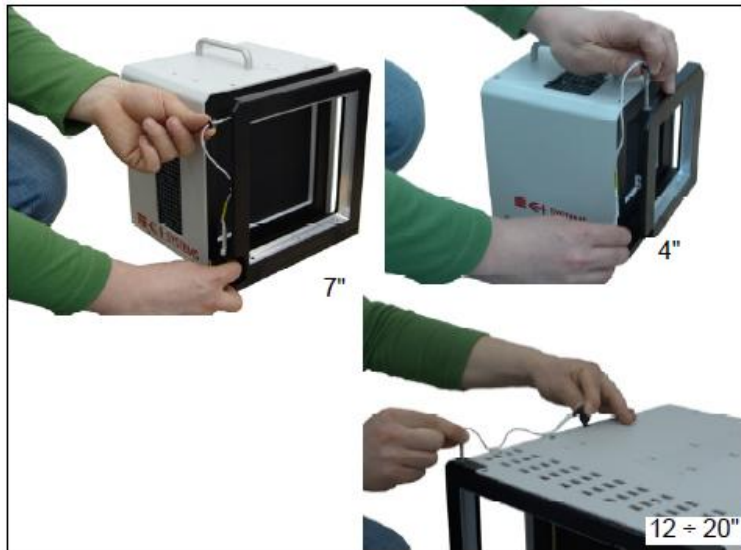
Do not use excessive force.

5.5.2 4 ÷ 20" Blackbody Heads Target Sensor Replacement

5.5.2.1 Removal

- Disconnect the Temperature Sensor connector from the Blackbody:

Figure 5-8: 4 ÷ 20" Blackbody Heads Target Sensor connector



- Hold the Temperature Sensor and carefully pull it out.

CAUTION

Do not pull the temperature Sensor by its cable.

NOTE

The Temperature sensor is secured by a built-in magnet.

Therefore, a reasonable force is required to separate the sensor from the Target holder.

- Remove the Temperature sensor and store it in its package to avoid damage.
- Clean the Sensor port using the supplied plastic straw - Insert it into the Sensor port, rotate it several times and remove it.

5.5.2.2 Installation

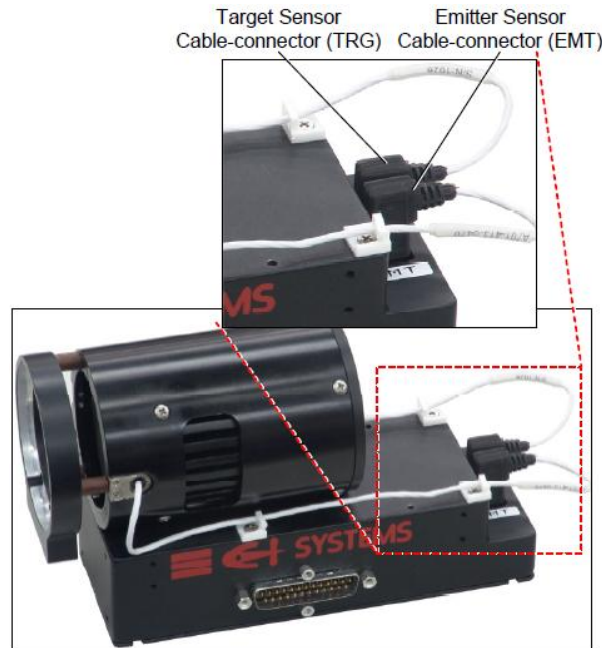
- Apply a thin layer of thermal grease (supplied with the kit) on the bottom section of the Temperature sensor (approx. 1cm on the free end of the sensor), using a toothpick (see [Figure 5-7](#)).
- Insert the Temperature sensor into the Target port, verify the magnet secures it.
- Connect the Temperature Sensor connector to the Blackbody (see [Figure 5-8](#)).

5.5.3 2" Blackbody Head Emitter Sensor Replacement

5.5.3.1 Removal

- Disconnect the Emitter Temperature Sensor cable-connector (marked **EMT**):

Figure 5-9: 2" Blackbody Head Emitter Sensor Replacement



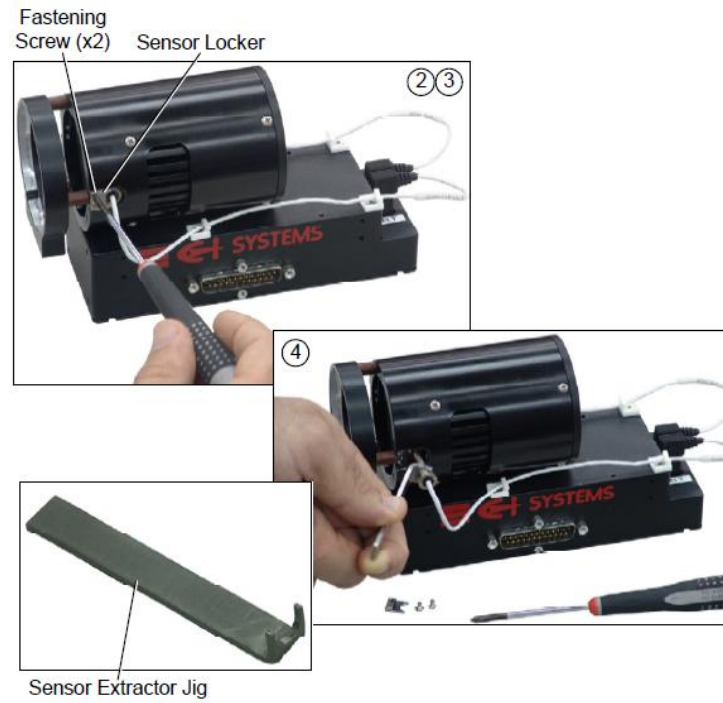
- Remove the screws securing the Sensor Locker to the Blackbody.
- Remove the Sensor Locker (see [Figure 5-10](#)).

NOTE

The inserted sensor is held in place by a built-in magnet; therefore, a reasonable force is required to separate the sensor from its port.

- Remove the Emitter Temperature Sensor using the Sensor Extractor Jig (supplied with the new sensors):

Figure 5-10: 2" Blackbody Head Emitter Sensor pulling out



- Store the sensor in its package to avoid damage.
- Clean the Sensor port using the supplied plastic straw - Insert it into the Sensor port, rotate it several times and remove it.

5.5.3.2 Installation

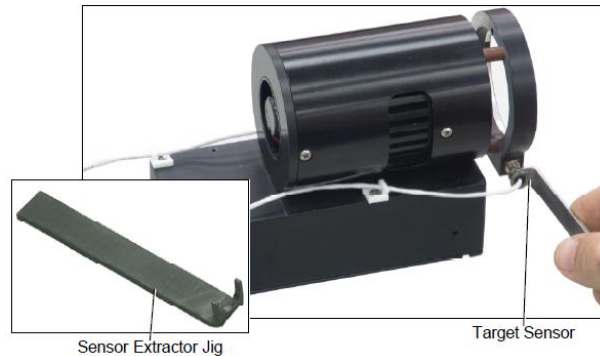
- Apply a thin layer of thermal grease (supplied with the kit) on the bottom section of the Temperature sensor (approx. 1cm on the free end of the sensor), using a toothpick (see [Figure 5-7](#)).
- Insert the Temperature sensor into the port until the magnet secures it (see [Figure 5-10](#)).
- Place the Sensor Locker and secure using the two screws (see [Figure 5-10](#)).
- Connect the Temperature Sensor cable-connector (marked 'EMT', see [Figure 5-9](#)).
- Arrange the Temperature sensor cable.

5.5.4 2" Blackbody Head Target Sensor Replacement

5.5.4.1 Removal

- Disconnect the Target Temperature Sensor cable connector (marked 'TRG', see [Figure 5-9](#)).
- Remove the Target Temperature Sensor using the Sensor Extractor Jig (supplied with the new sensors):

Figure 5-11: 2" Blackbody Head Target Sensor pulling out



NOTE

The Temperature sensor is held in place by a built-in magnet; therefore, a reasonable force is required to separate the sensor from its port.

- Store the Temperature sensor in its package to avoid damage.
- Clean the Sensor port using the supplied plastic straw - Insert it into the Sensor port, rotate it several times and remove it.

5.5.4.2 Installation

- Apply a thin layer of thermal grease (supplied with the kit) on the bottom section of the Temperature sensor (approx. 1cm on the free end of the sensor), using a toothpick (see [Figure 5-7](#)).
- Insert the Temperature sensor into the port until the magnet secures it (see [Figure 5-10](#)).
- Connect the Temperature Sensor cable-connector (marked 'TRG', see [Figure 5-9](#)).
- Arrange the Temperature sensor cable.

5.5.5 Updating the Calibration Data

The new Calibration data is stored in a USB mass storage device, provided with the new calibrated sensor.

5.5.5.1 Upgrade screen

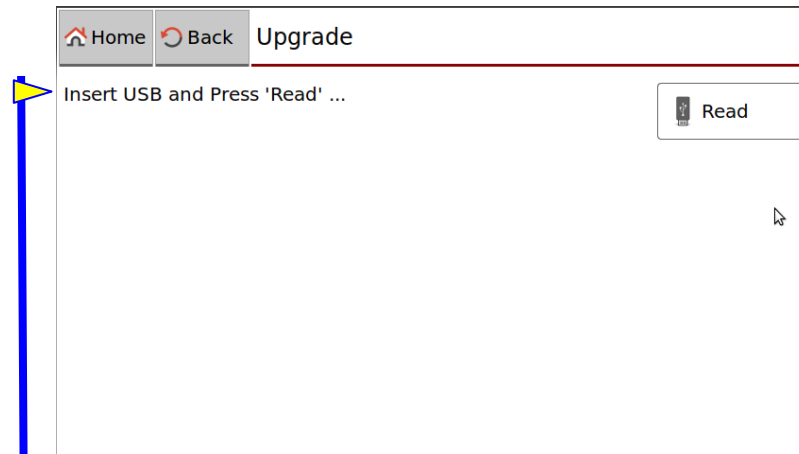
To invoke the Upgrade screen press the  button in the 'Menu' screen (see [Section 4.3.7](#)).

The instructions to the Operator and the status of the procedure are displayed on the screen.

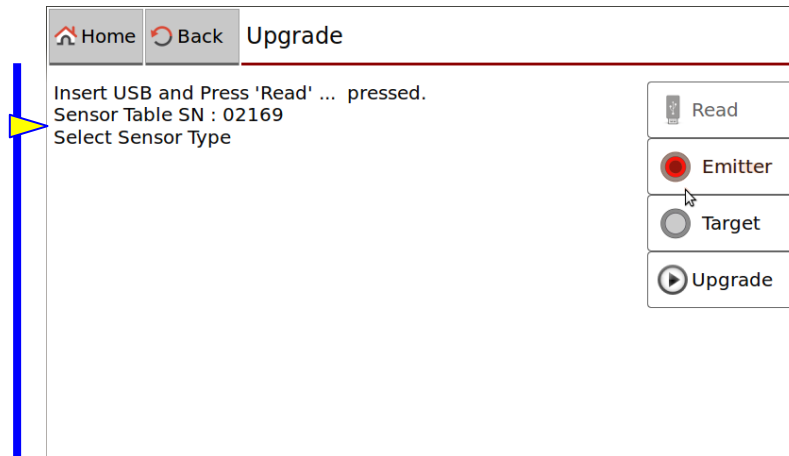
5.5.5.1.1 Loading Calibration data screen

The following paragraph is a step-by-step description of the procedure to load a new calibration data.

Figure 5-12: Upgrade screen - Loading Calibration data screen (1/5)



- After pressing the  button in the 'Upgrade screen, the content of the USB Mass Storage Device that was inserted to the USB jack is checked.

Figure 5-13: Upgrade screen - Loading Calibration data screen (2/5)

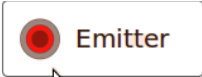
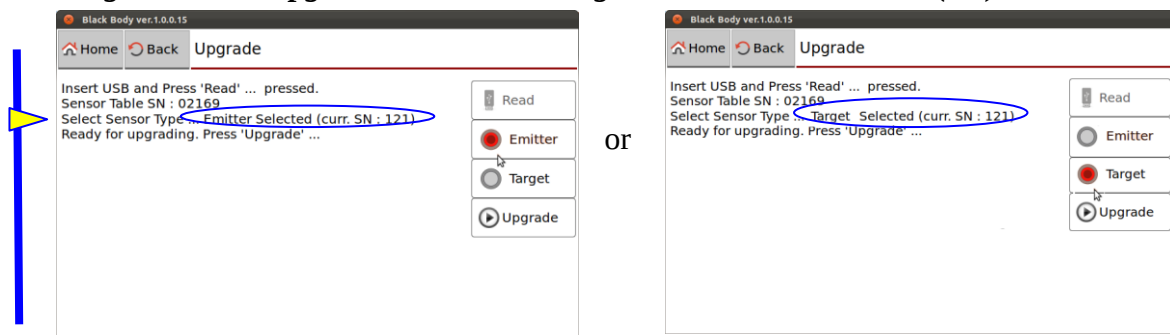
- In case that a new calibration data is detected – the S/N of the new sensor is displayed.
- By pressing one of the sensor type buttons:  or  the defines if this calibration data is related to the new Emitter or Target sensor (default is emitter).

Figure 5-14: Upgrade screen - Loading Calibration data screen (3/5)

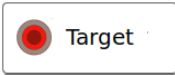
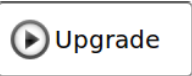
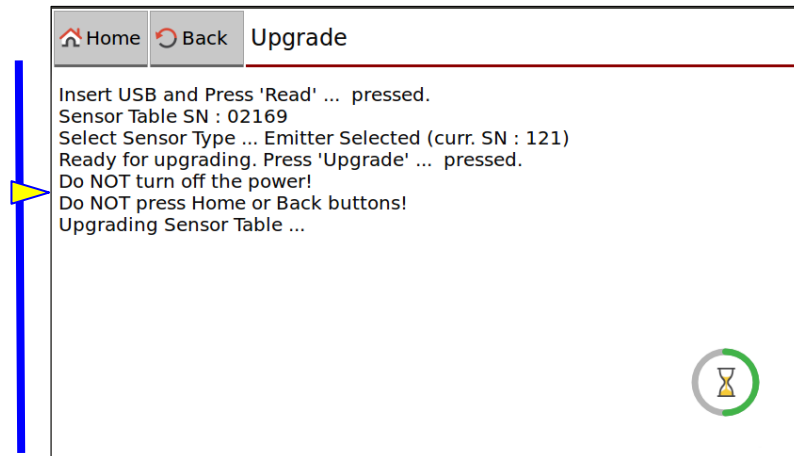
- The button of the selected sensor is marked  or , and the selected type is displayed in the instructions.
- Pressing the  button initiates the update procedure – the new Calibration Data will replace the old one.

Figure 5-15: Upgrade screen - Loading Calibration data screen (4/5)

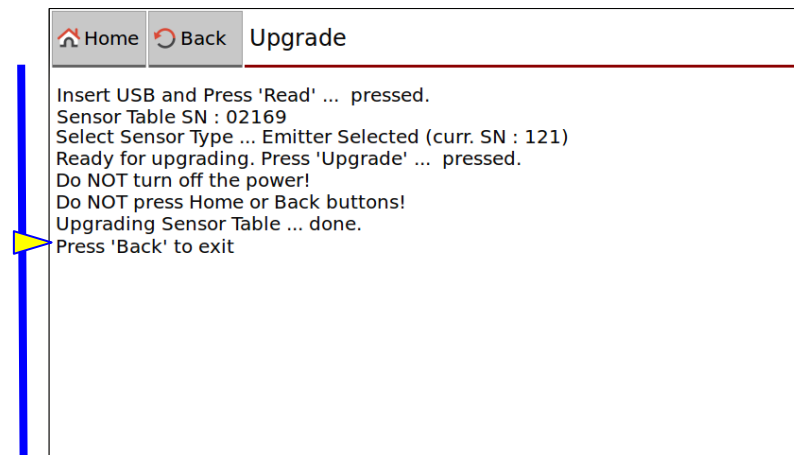
- During this procedure, the progress is indicated by  , along with instructions to the operator.

NOTE

During the update procedure:

Do not power off the SR-800N or press any button.

Failure to comply will result with corrupted calibration data.

Figure 5-16: Upgrade screen - Loading Calibration data screen (5/5)

- If the procedure ended successfully, the progress indicator disappears, and the screen indicates "Upgrading Sensor Table ... done"
- Press the 'Back' button to close the screen and return to the 'Menu' screen.

5.6 Updating the Software Version

This following procedure enables to upgrade the software version of the SR-800N control cards (SBC, HCC and MCC).

The new Software Version is stored in a USB mass storage device, provided from CI Systems.

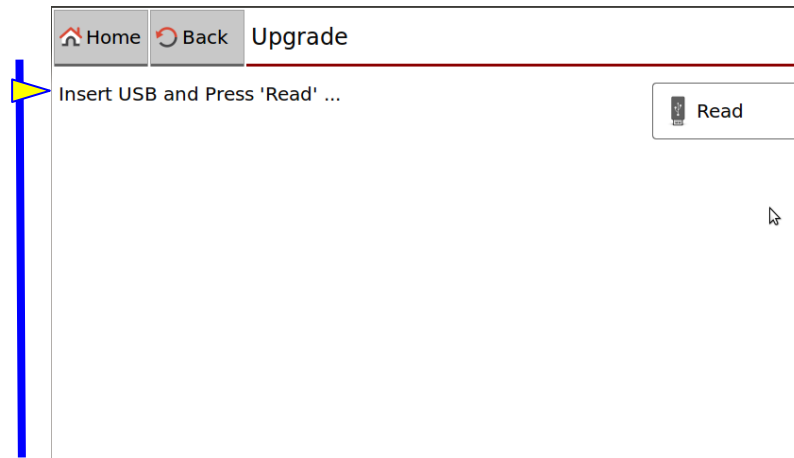
5.6.1 Upgrade screen

To invoke the Upgrade screen press the  button in the 'Menu' screen (see [Section 4.3.7](#)).

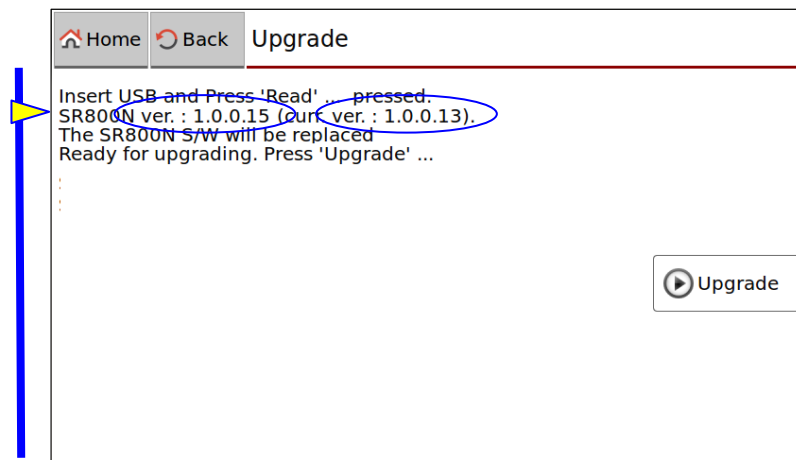
The instructions to the Operator and the status of the procedure are displayed on the screen.

5.6.1.1 Upgrade Software screen

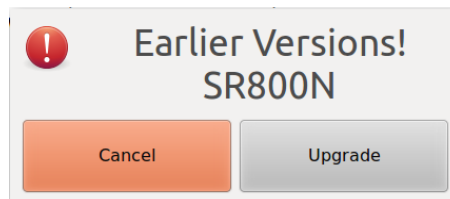
Figure 5-17: Upgrade screen - Upgrade Software screen (1/4)



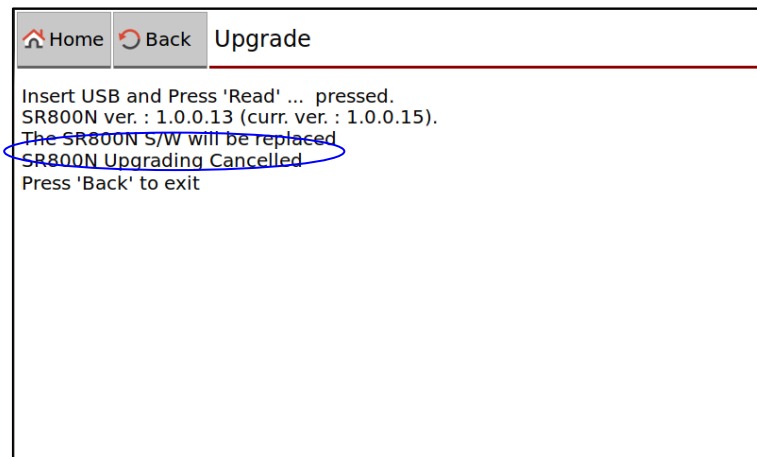
- After pressing the  in the Upgrade Software screen, the content of the USB Mass Storage Device that was inserted to the USB jack is checked.

Figure 5-18: Upgrade screen - Upgrade Software screen (2/4)

- If a file with software version is identified the Upgrade Software screen displays the version of the new file, and displays the version of the current installed software.
- If the new file has higher lower version number, then a warning pop-up message will be displayed

Figure 5-19: Upgrade screen - Upgrade Software screen Warning

- Confirm the earlier version by pressing the 'Upgrade' button or quit the procedure by pressing the 'Cancel' button.

Figure 5-20: Upgrade screen - Upgrade Software screen Canceled

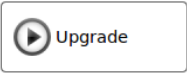
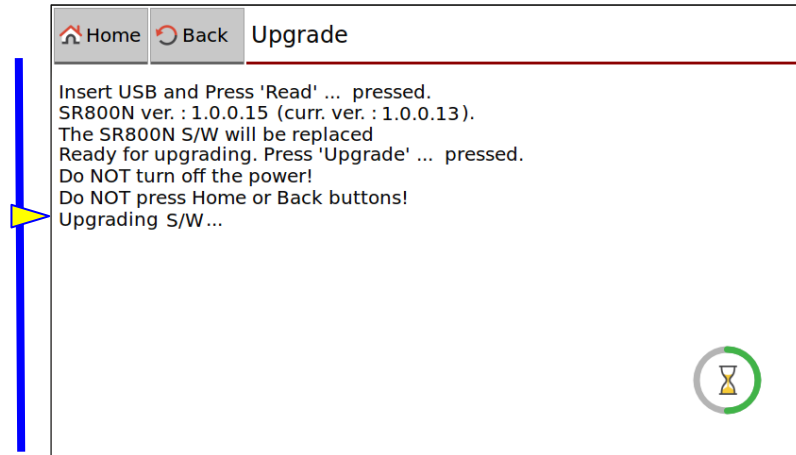
- Pressing the  button initiates the software upgrade procedure – the new version will replace the old one.

Figure 5-21: Upgrade screen - Upgrade Software screen (3/4)



- During this procedure, the progress is indicated by  , along with instructions to the operator.

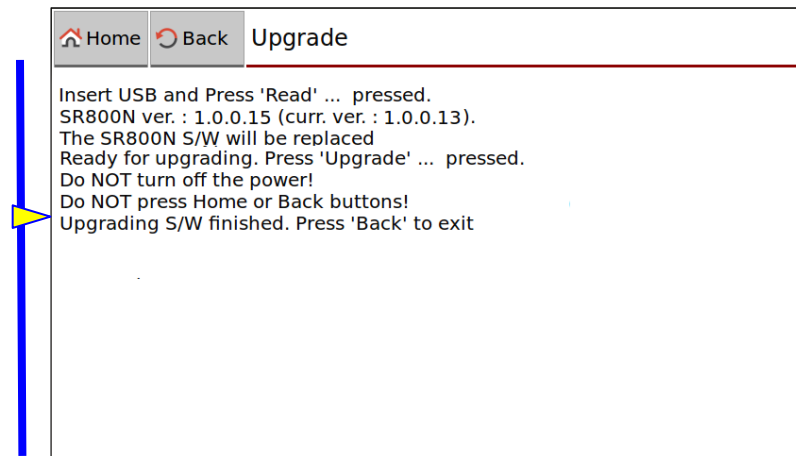
NOTE

During the update procedure -

Do not power off the SR-800N or press any buttons.

Failure to comply will result with corrupted calibration data.

Figure 5-22: Upgrade screen - Upgrade Software screen (4/4)



- At the end of the procedure the progress indicator disappears, and the screen indicates "Upgrading S/W finished".
- Press the 'Back' button to close the screen.
- The application will now restart.

6. Communication Channels

This chapter defines the communications channels of the SR-800N ControlMaster. These ports can be used to communicate with a Host Computer or other devices.

6.1 Communication Ports description

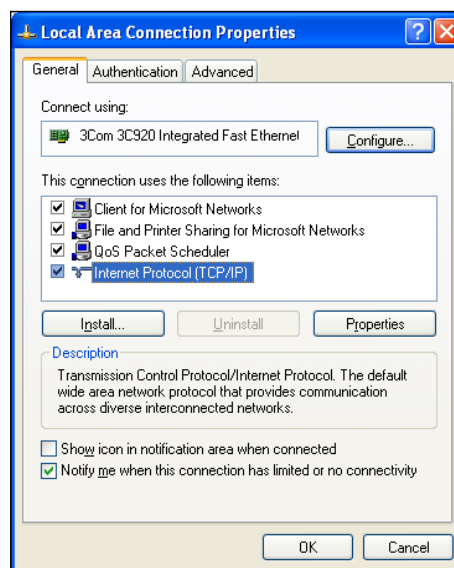
6.1.1 RS232 serial port

- Baud rate: 115200 bps (default) or 9600 bps (see [Section 4.4.1.3](#) for communication setting description)
- Number of Data bits – 8
- Number of Stop bits -1
- No parity.

6.1.2 Ethernet

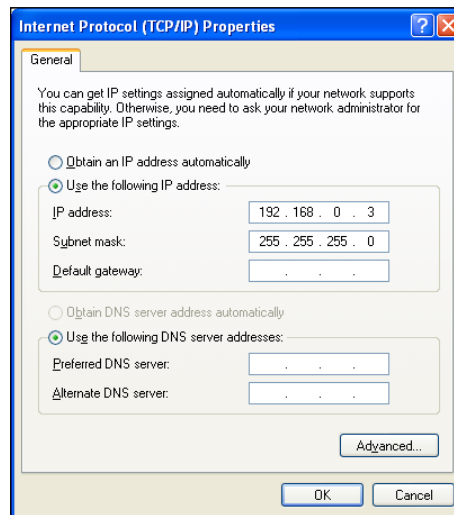
- Ethernet Port: 10/100 BaseT
- Can operate either at dynamic using DNS, or static using fixed address. (see [Section 4.4.1.3](#) for communication setting description)
- When connecting a Host directly using fixed address, verify to set the Host adapter to correct address
- To setup the Host Computer for Direct Link (in cases where there is no in-house network and/or for security reasons):
 - From the computer Start menu select Network Connections (Start → Settings → Network Connections → Local Area Connection) The following window opens:

Figure 6-1: Setting Up the Host Computer for Direct Link



- Select the **Internet Protocol (TCP/IP)** and click '**Properties**'. The **Internet Protocol (TCP/IP) Properties** window opens.

Figure 6-2: Setting Up the Internet Protocol (TCP/IP)



- Enter an IP address – similar to the one displayed in the ControlMaster's 'Info' screen, but with different last number.
- Enter subnet mask – identical to the one displayed in the ControlMaster's 'Info' screen.
- Click 'OK' to complete close the form.

6.2 Remote Operation

6.2.1 Overview

Using one of the communication ports, the SR-800N can be operated remotely from a Host Computer.

The remote operation capability enables to integrate the SR-800N in an Automatic Test Equipment (ATE).

6.2.2 Communication Protocol

Refer to '*SR-800N Communication Protocol description*' document for detailed description.

Although the communication protocol of the SR-800N is different than the one that is used in the SR-800/SR-800R/SR80 systems, the SR-800N can communicate with external devices using this version of protocol.

For new designs, it is recommended to use the enhanced new communication protocol.

6.2.3 Remote Operation Application

The SR-800N is provided with a remote operation application.

This application runs on the host PC, connected to the SR-800N using one of the optional communication channels (see [Section 6.1](#))

Refer to '*SR-800N Remote Operation Application Manual*' document for detailed installation and operation information.

7. Appendix A – Targets Wheel

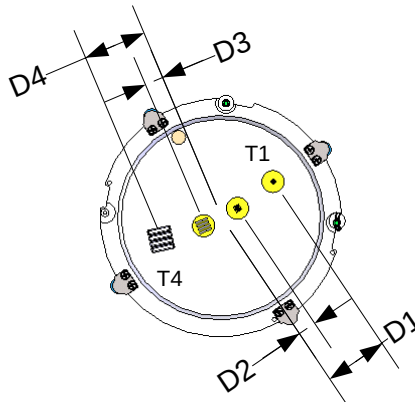
This chapter describes the construction of the Targets Wheel of the SR-800N and the procedure to define it using the ControlMaster.

7.1 General

7.1.1 Targets Plate construction

- The Targets Plate is a round plate that holds several targets, with unique serial number (marked on the plate).
- Each Targets Plate contains up to 4 different targets.
- Each Target has the following parameters (not including its optical ones):
 - Unique serial number (T1, T2, etc.).
 - Name (free textual field).
 - The serial number of the Targets Plate that contains it.
 - Relative placement, with respect to the center of the Targets Plate (can be positive or negative), measured in 'Steps' (D1, D2..).
- The Targets Plate is equipped with small tabs that secure it its position when they are engaged to the magnets that are installed in the Targets Wheel.
- The Shape of the Targets Plate ensures its orientation when it is installed in the Targets Wheel.

Figure 7-1: Targets Plate construction (for reference)



7.1.2 Targets wheel construction

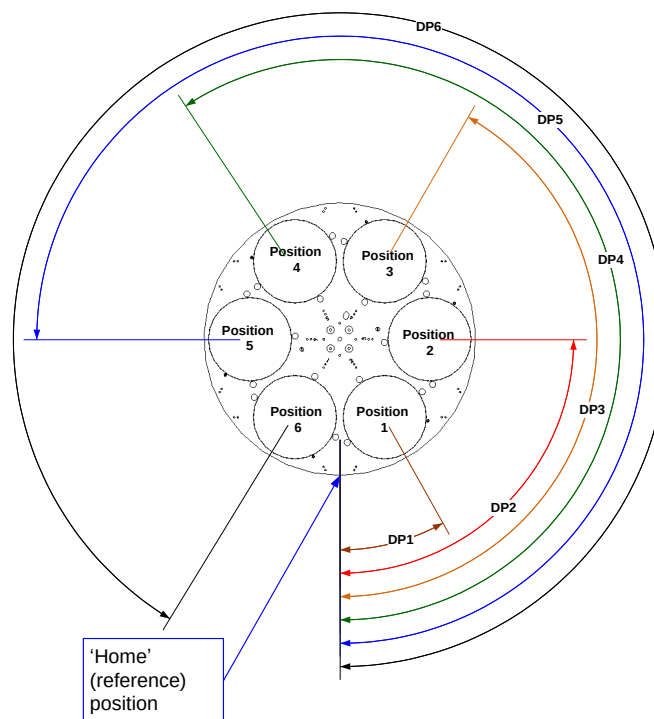
- The Targets wheel is a round plate, with holes (marked as 'Positions').
- Each 'Position' can accommodate a Targets Plate.
- A set of magnets is installed around each hole in Targets Wheel to hold the Targets Plate in its place.

- The Targets wheel is connected to a step-motor and motion controller (located in the ControlMaster).
- The Targets wheel uses a slotted optical switch to indicate its reference position (marked as 'Home').
- The Targets wheel has the following parameters (Factory set):
 - Total number of 'Positions'.
 - Absolute placement of each 'Position', with respect to the 'Home' (positive only), measured in 'Steps' (DP1, DP2...).

NOTE

When placing a Targets Plate in a 'Position',
the placement of its center accepts the placement value
of the Position.

Figure 7-2: Targets wheel construction(for reference)



7.2 Operation

7.2.1 Positioning a Target

- Follow the instructions in [Section 4.3.8](#) to select a Target.
- The motion controller in the ControlMaster calculates the absolute placement of the target:

$$\text{Target}_{\text{ABS}} = \text{Target}_{\text{REL}} + \text{Target_Plate}_{\text{ABS}}$$

Where:

Target_{ABS}: The absolute placement of the selected target, with respect to the 'Home' of the Targets Wheel.

Target_{REL}: The relative placement of the selected target, with respect to the center of its Targets Plate (see D1, D2.. in [Figure 7-1](#)).

Target_Plate_{ABS}: The absolute placement of the center of the selected Targets Plate, with respect to the 'Home' of the Targets Wheel.

This parameter is actually the placement of the 'Position' of the Targets Wheel, at which this Target Plate is placed (see DP1, DP2.. in [Figure 7-2](#)).

- The motion controller counts the placements activities and compare it to the 'Homing Cycle' value). If the count is less than the 'Homing Cycle' or 'Homing Cycle' is set to '0' than homing cycle is not needed, otherwise homing cycle is required.
- If homing cycle is not needed, the motion controller compares the current placement of the Targets Wheel with the required new one and rotates it in the shortest way (CW or CCW).

NOTE

The default behavior of the Targets Wheel is

not to cross the 'Home' position.

- If homing cycle is required, the motion controller rotates Targets Wheel to the 'Home' position and then to the required new placement.
- The default 'Homing cycle' value is 10, however, the operator can modify it using the 'Axis setting' screen (see [Section 4.4.3](#)).

7.3 Calibration

The SR-800N enables to perform the following activities

- Managing the Axis (Targets Wheel) parameters.
- Calibrating the 'Home' (reference) position of the Targets Wheel.
- Managing the Targets Plates placement in the Targets Wheel.
- Tuning Target Relative placement.
- Managing Target information.

7.3.1 Managing the Axis (Targets Wheel) parameters

- The operator can modify the 'Homing Cycle' parameter (see [Section 4.4.3.4.1](#)), to change or disable (by setting it to '0') the frequency of this function.
- The operator can set or modify the name of the Axis (Target Wheel) for a meaningful value (see [Section 4.4.3.4.3](#)). This name is displayed in the activating button in the main screen and at the header of each screen related to this Axis.
- The operator can control the step size of the motor that drive the Target Wheel (see [Section 4.4.3.4.3](#)).
This function should be carried out carefully because it affects the targets placement calibration and positioning accuracy.

7.3.1 Calibrating the 'Home' (reference) position of the Targets Wheel

- The operator can use the 'Direct Operation' (see [Sections 4.4.3.4.4 and 4.4.5](#)) to rotate the target wheel to its 'Home' position, for example, according to the center of its 'Position 1' (see [Figure 7-2](#)).
- In order to set this position as its 'Home', the operator should set the 'Home Offset' parameter (see [Section 4.4.3.4.2](#)).

7.3.2 Managing a single Target information

- The operator can use the Axis Item Configuration screen (see [Section 4.4.4](#)) to view and modify all the targets that are currently defined.
- Each Target shall be related to a Target Plate, otherwise the ControlMaster will not be able to calculate its absolute placement of the.

7.3.3 Managing the Targets Plates placement

- The operator can use the Targets Plate placement screen (see [Section 4.4.3.3](#)) to view all the targets that are currently defined in the Target Wheel.
- The operator can select which Target Plate is placed at each Position of the Target Wheel.
- There is an option to define a Position as 'Empty', so that the Targets Wheel will never stop at this 'Position'.

7.3.4 Tuning Target Relative placement

- The operator can use the Tuning screen (see [Section 4.4.4.1](#)) test and fine tune the relative position (see D1, D2.. in [Figure 7-1](#)) of a target.

7.3.5 Testing Target absolute placement

- The operator can use the forced 'homing', available from the 'Axis Setting' screen (see [Figure 4-22](#)) or 'Target Tuning' screen (see [Figure 4-29](#)), to the absolute placement of a target. This method is preferred because it eliminates any accumulative errors in the positioning.

NOTE

The placement algorithm works in 'Open loop' method
(only counting step commands to the Step-Motor).
Therefore the homing cycle clears the accumulative
placement error due to skipped steps.

8. Index

This chapter provides, in alphabetical order, a list of keywords to be found throughout this manual.

A

Abbreviations · xi

Absolute Mode · 2-5

Advanced Configuration

- Power-Up Preset Setting · 4-45
- Restore the factory settings · 4-47
- Stabilization Window Setting · 4-46
- Temp. precision setting · 4-45
- Turning On/Off the Axes control · 4-47
- Turning On/Off the BB Head · 4-46
- Turning On/Off the Buzzer · 4-47

Applicable Documents · xi

Axis Setting

- Direct Operation · 4-63

Axis Setting

- Axis Configuration · 4-55
- Axis Name · 4-56
- Axis Placement · 4-53
- Enable/Disable Direct Operation · 4-57
- Home Offset Value · 4-56
- Homing · 4-53
- Homing Cycle Value · 4-56
- Step Size Selection · 4-57
- Targets Plate Selection · 4-54
- Turning On/Off · 4-52

B

Blackbody Configuration

- Mode of Operation · 4-43
- Set Point Progress setting · 4-43
- Temp. Measurement Units · 4-43
- Up/Down Step Size setting · 4-43

Blackbody Head · 2-4

- Absolute Mode · 2-5
- Differential Mode · 2-5
- Julabo Circulation System · 2-7
- Structure · 2-6

C

Calibration · 5-64

Calibration Methods

- Removable temperature sensor · 5-70
- standalone calibration instrument · 5-70

Calibration Procedure

- Update the Calibration Data · 5-80

Cleaning · 5-64

Communication Channels · 6-1

Communication ports

- Ethernet · 6-1
- RS232 · 6-1

Configuration

- Advanced Configuration · 4-44
- Axis Item Configuration · 4-58

Axis Setting · 4-52
Blackbody Configuration · 4-42
Communication Setting · 4-48
Main Menu · 4-41
MCC Configuration · 4-49
Preset 'Axis 1' · 4-51
Preset Configuration · 4-50
Preset Mode selection · 4-51
Preset Name Editing · 4-51
Preset 'Set Point' · 4-51
Procedures · 4-41
Target Relative Position Tuning · 4-62
Target Type Selection · 4-60
ControlMaster
 Components · 2-8
 Connectors · 2-9
 Controls & Indicators · 2-8
 I/O Connector - Encoder Feedback Option ·
 2-12
 I/O Connector - Galvanic Isolation Option · 2-
 12
 I/O Connector - TTL Option · 2-11
 LAN Jack · 2-11
 RS232 Connector · 2-10
 USB Jack · 2-10
ControlMaster · 2-8
Copyright · ii

D

Description · 2-1
 Blackbody Head · 2-4
 ControlMaster · 2-8
 SR-800N Block Diagram · 2-1

SR-800N Cables set · 2-14
SR-800N Main Components · 2-2
SR-800N Main Options · 2-15
Differential Mode · 2-5

E

Effective Pages · iv

F

Features · 1-1, 1-2
Fuse Replacement · 5-69

G

General Information · 1-1
 ControlMaster Electrical specification · 1-3
 ControlMaster Mechanical specification · 1-3
 Features · 1-2
 Overview · 1-1
 Specifications · 1-3
 SR-800N performance · 1-3

H

Handling & Installation · 3-16

I

Installation · 3-16
 19 · 3-17
 Desktop (stand alone) · 3-17
 Julabo pipes · 3-21
 LT Blackbody Head Cables Set · 3-22
 LT Blackbody Head Hood · 3-23
 LT Blackbody Head pipes · 3-20
 LT Blackbody Head Unpacking · 3-19

Unpacking · 3-16

Introduction · 1-1

L

List of Figures · viii

List of Tables · viii

LT Blackbody Head

Safety Precautions · 3-19

LT Blackbody Head Installation · 3-19

M

Maintenance · 5-64

Calibration methods · 5-70

Calibration Procedure · 5-80

Fuse Replacement · 5-69

Temperature Sensors replacement · 5-72

Troubleshooting · 5-65

Updating the Software Version · 5-83

O

Operation · 4-26

Absolute Mode Screen · 4-31

Differential Mode Screen · 4-32

Failures list · 5-66

Filter Selection · 4-35

General Controls · 4-28

General Data Entry Screens · 4-29

General navigation buttons · 4-27

I/O Control · 4-37

Incremental Temperature Setting · 4-33

Info · 4-38

LT Blackbody Head Operation · 4-40

LT Blackbody Head Power-On · 4-40

LT Blackbody head set Temperature · 4-40

Main Menu · 4-41

Manual Temperature Setting · 4-33

Powering-ON · 4-26

Preset selection · 4-34

Procedures · 4-27

Restore the factory settings · 4-47

Selecting an Axis item · 4-35

Setting the Temperature · 4-33

Source Selection · 4-35

'Stable Temperature' Indication · 4-33

Target Selection · 4-35

Temperature Resolution · 4-33

Turning-Off · 4-26

Overview · 1-1

P

Preface · iii

Preset functions · 4-34

Preventive/Scheduled Maintenance

Calibration · 5-64

Cleaning · 5-64

Visual Inspection · 5-64

Preventive/Scheduled Maintenance Tasks · 5-64

R

Remote Operation · 6-3

Communication Operation Application · 6-3

Communication Protocol · 6-3

Removal · 3-24

19 · 3-24

Desktop · 3-24

LT Blackbody Head · 3-24

LT Blackbody Head Disconnecting the pipes · 3-25
LT Blackbody Head Fluid Draining · 3-24
LT Blackbody Head Hood · 3-25
LT Blackbody Head re-packing · 3-25
Restore the factory settings · 4-47

S

Safety Summary · xii

Screens

Absolute Mode · 4-31
Advanced Configuration · 4-44
Alphanumeric entry · 4-30
Axis Configuration · 4-55
Axis Item Configuration · 4-58
Axis Placement · 4-53
Axis Setting · 4-52
Blackbody Configuration · 4-42
Communication Setting · 4-48
Differential Mode · 4-32
Direct Operation · 4-63
Failures · 5-66
Filter Selection · 4-35
General Controls · 4-28
I/O Control · 4-37
Info · 4-38
Main Menu · 4-41
MCC Configuration · 4-49
Mode selection · 4-51
Navigation buttons · 4-27
Numeric entry · 4-29
Power-Up Preset setting · 4-45
Preset Configuration · 4-50
Preset selection · 4-34
Set Point Adjustment · 4-33
Set Point Progress setting · 4-43
Source Selection · 4-35
Step Size Selection · 4-57
Target Selection · 4-35
Target Tuning · 4-62
Target Type Selection · 4-60
Targets Plate Selection · 4-54
Temp. precision setting · 4-45
Temperature Resolution · 4-33
'Unsaved Changes' Warning · 4-27
Up/Down Step Size setting · 4-43
Upgrade calibration data · 5-80
Upgrade S/W · 5-83
Sensor Replacement
 2 · 5-77, 5-79
 4 ÷ 20 · 5-72, 5-76
Setting the Temperature · 4-33
Specifications · 1-3
SR-800N Cables set · 2-14
SR-800N Main Options · 2-15
 Emissivity Compensation · 2-15
 Encoder feedback · 2-15
 Extended Temperature Range · 2-15
 High Emissivity Emitter · 2-15
 High Uniformity · 2-15
 Low Temperature · 2-15
 Motorized Axes · 2-15
 Special I/Os · 2-15
 Wide Temperature Range · 2-15
Storage · 3-16

T

Table of Contents · v

Targets Wheel · 7-1

 Calibration · 7-4

 Operation · 7-3

 Plate construction · 7-1

 Wheel construction · 7-1

Technical Support · ii

Temperature sensors Calibration methods · 5-70

Temperature Sensors Maintenance · 5-72

Transportation · 3-16

Troubleshooting · 5-65

 Blackbody Failures messages · 5-67

 ControlMaster Failures messages · 5-68

 Failure list · 5-66

 General Failures · 5-65

V

Visual Inspection · 5-64

W

Warning Messages

 Restore the factory settings · 4-47